

E-companion for Enhancing Safety Signaling: Integrating Clinical Trials and Post-Marketing Adverse Event Reports

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Section A: Preliminaries on normal distributions

A.1. Bivariate normal distributions

Consider a standard bivariate normal distribution,

$$X \sim \mathcal{N}\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & \rho_{ab} \\ \rho_{ab} & 1 \end{bmatrix}\right).$$

Its cdf is given by

$$\begin{aligned} \tilde{\Phi}(x_a, x_b; \rho_{ab}) &:= \mathbb{P}\left(X \leq \begin{bmatrix} x_a \\ x_b \end{bmatrix}\right) \\ &= \frac{1}{2\pi\sqrt{1-\rho_{ab}^2}} \int_{-\infty}^{x_a} \int_{-\infty}^{x_b} \exp\left\{\frac{-1}{2(1-\rho_{ab}^2)} \left[u^2 - 2\rho_{ab}uv + v^2\right]\right\} dv du. \end{aligned} \quad (\text{EC.1})$$

Given $\tilde{\Phi}(x_a, x_b; \rho_{ab})$, the partial derivative with respect to x_i , $i \in \{a, b\}$, is given by

$$\frac{\partial \tilde{\Phi}}{\partial x_i} = \phi(x_i) \Phi\left(\frac{x_j - \rho_{ab}x_i}{\sqrt{1-\rho_{ab}^2}}\right), \quad (\text{EC.2})$$

and the partial with respect to the correlation (Drezner and Wesolowsky (1990), Eq. (4)) is

$$\begin{aligned} \frac{\partial \tilde{\Phi}}{\partial \rho_{ab}} &= \frac{1}{2\pi\sqrt{1-\rho_{ab}^2}} \exp\left\{\frac{-(x_a^2 - 2\rho_{ab}x_ax_b + x_b^2)}{2(1-\rho_{ab}^2)}\right\} \\ &= \frac{1}{\sqrt{1-\rho_{ab}^2}} \phi(x_b) \phi\left(\frac{\rho_{ab}x_b - x_a}{\sqrt{1-\rho_{ab}^2}}\right) = \frac{1}{\sqrt{1-\rho_{ab}^2}} \phi(x_a) \phi\left(\frac{\rho_{ab}x_a - x_b}{\sqrt{1-\rho_{ab}^2}}\right). \end{aligned} \quad (\text{EC.3})$$

A.2. Gaussian smoothing of functions

DEFINITION EC.1 (GAUSSIAN SMOOTHING). For $\sigma > 0$, the *Gaussian smoothing* $f_\sigma(x)$ of a function $f(x)$ is defined as

$$f_\sigma(x) := \int_{-\infty}^{\infty} f(x + \sigma z) \phi(z) dz.$$

In brief, Gaussian smoothing returns a smoothed version of a function by averaging nearby function values. The bandwidth $\sigma > 0$ captures the degree of smoothness (larger σ implies more smoothness, while small σ returns something very close to the original function). This concept of Gaussian smoothing of a function is defined and generalized to multiple dimensions in Nesterov and Spokoiny (2017). They establish a number of properties: (i) $f_\sigma(x)$ is continuous and differentiable, (ii) $f'_\sigma(x)$ is continuous, and (iii) $f'_\sigma(x)$ takes a convenient functional form (see Eq. (21)). Specifically,

$$f'_\sigma(x) = \int f(x + \sigma z) \frac{z\phi(z)}{\sigma} dz. \quad (\text{EC.4})$$

Section B: Model preliminaries

B.1. Loss functions and expected optimal cost-to-go

We first present an equivalent expression for the loss functions.

LEMMA EC.1. *Consider the estimate of the true effect size at time $l \in [A, S]$, $\mu_2 | \hat{\mu} \sim \mathcal{N}(\hat{\mu}_2^l, (\sigma_2^l)^2)$, and some monotone type I and type II error costs $T_I^l(\mu_2)$ and $T_{II}^l(\mu_2)$, such that $T_I^l(\mu_2)$ ($T_{II}^l(\mu_2)$) takes value 0 for $\mu_2 > 0$ ($\mu_2 < 0$), and assigns some cost to errors $T_I^l(-\infty) > 0$ ($T_{II}^l(\infty) > 0$) otherwise. Both $T_I^l(\mu_2)$ and $T_{II}^l(\mu_2)$ grow as $o(e^{\epsilon \mu_2^2})$ for any $\epsilon > 0$ as $\mu_2 \rightarrow -\infty$ and $\mu_2 \rightarrow \infty$, respectively. Then, the expected loss of (not) taking action at time l (labeling or signaling) can be written as*

$$\mathcal{L}_l(\hat{\mu}_2^l, \sigma_2^l, 1) := \mathbb{E}_{\mu_2 | \hat{\mu} \sim \mathcal{N}(\hat{\mu}_2^l, (\sigma_2^l)^2)}[T_I^l(\mu_2)] = \int_{-\infty}^{\infty} \Phi\left(\frac{x - \hat{\mu}_2^l}{\sigma_2^l}\right) (-dT_I^l(x)), \text{ and} \quad (\text{EC.5})$$

$$\mathcal{L}_l(\hat{\mu}_2^l, \sigma_2^l, 0) := \mathbb{E}_{\mu_2 | \hat{\mu} \sim \mathcal{N}(\hat{\mu}_2^l, (\sigma_2^l)^2)}[T_{II}^l(\mu_2)] = \int_{-\infty}^{\infty} \Phi^C\left(\frac{x - \hat{\mu}_2^l}{\sigma_2^l}\right) dT_{II}^l(x). \quad (\text{EC.6})$$

Proof. To derive the result, we express the expected loss using Riemann–Stieltjes integration over the cdf of the random variable $\mu_2 | \hat{\mu}$ and then perform integration by parts:

$$\begin{aligned} \mathcal{L}_l(\hat{\mu}_2^l, \sigma_2^l, 1) &:= \mathbb{E}_{\mu_2 | \hat{\mu} \sim \mathcal{N}(\hat{\mu}_2^l, (\sigma_2^l)^2)}[T_I^l(\mu_2)] \\ &= \lim_{M \rightarrow \infty} \int_{-M}^M T_I^l(x) d\Phi\left(\frac{x - \hat{\mu}_2^l}{\sigma_2^l}\right) \\ &= \lim_{M \rightarrow \infty} \underbrace{T_I^l(M)}_{=0} \Phi\left(\frac{M - \hat{\mu}_2^l}{\sigma_2^l}\right) - \underbrace{T_I^l(-M) \Phi\left(\frac{-M - \hat{\mu}_2^l}{\sigma_2^l}\right)}_{\rightarrow 0 \text{ (} T_I^l(-M) = o(e^{\epsilon M^2}) \text{ for any } \epsilon > 0)} - \int_{-M}^M \Phi\left(\frac{x - \hat{\mu}_2^l}{\sigma_2^l}\right) dT_I^l(x) \\ &= \int_{-\infty}^{\infty} \Phi\left(\frac{x - \hat{\mu}_2^l}{\sigma_2^l}\right) (-dT_I^l(x)). \end{aligned}$$

The result for the no-taking action follows the same argument, though we add a step in which we add and subtract the term $T_{II}^l(M) = \int_{-M}^M dT_{II}^l(x)$:

$$\begin{aligned} \mathcal{L}_l(\hat{\mu}_2^l, \sigma_2^l, 0) &:= \mathbb{E}_{\mu_2 | \hat{\mu} \sim \mathcal{N}(\hat{\mu}_2^l, (\sigma_2^l)^2)}[T_{II}^l(\mu_2)] \\ &= \lim_{M \rightarrow \infty} \int_{-M}^M T_{II}^l(x) d\Phi\left(\frac{x - \hat{\mu}_2^l}{\sigma_2^l}\right) \\ &= \lim_{M \rightarrow \infty} T_{II}^l(M) \Phi\left(\frac{M - \hat{\mu}_2^l}{\sigma_2^l}\right) - \underbrace{T_{II}^l(-M) \Phi\left(\frac{-M - \hat{\mu}_2^l}{\sigma_2^l}\right)}_{=0} - \int_{-M}^M \Phi\left(\frac{x - \hat{\mu}_2^l}{\sigma_2^l}\right) dT_{II}^l(x) \\ &= \lim_{M \rightarrow \infty} \underbrace{-T_{II}^l(M) \Phi^C\left(\frac{M - \hat{\mu}_2^l}{\sigma_2^l}\right)}_{\rightarrow 0 \text{ (} T_{II}^l(M) = o(e^{\epsilon M^2}) \text{ for any } \epsilon > 0)} + \int_{-M}^M \Phi^C\left(\frac{x - \hat{\mu}_2^l}{\sigma_2^l}\right) dT_{II}^l(x) = \int_{-\infty}^{\infty} \Phi^C\left(\frac{x - \hat{\mu}_2^l}{\sigma_2^l}\right) dT_{II}^l(x). \quad \square \end{aligned}$$

OBSERVATION 1. *Consider the case $l = S$ in Lemma EC.1, so we have that $\mu_2 | \hat{\mu} \sim \mathcal{N}(\hat{\mu}_2^S, (\sigma_2^S)^2)$. Noting that the expected type I and II error costs are strictly monotone decreasing and increasing, respectively, in $\hat{\mu}_2^S$, then there is a unique t_S^* at which $\mathcal{L}_S(t_S^*, \sigma_2^S, 0) = \mathcal{L}_S(t_S^*, \sigma_2^S, 1)$. That is, t_S^* satisfies the following equation*

$$\int_{-\infty}^{\infty} dT_{II}^S(x) - \int_{-\infty}^{\infty} \Phi\left(\frac{x - t_S^*}{\sigma_2^S}\right) (dT_{II}^S(x) - dT_I^S(x)) = 0. \quad (\text{EC.7})$$

We next provide an alternative formulation of the cost-to-go V_S^* that relies on jointly modeling the distribution of the true effect size μ_2 and the posterior mean $\hat{\mu}_2^S$.

LEMMA EC.2. Consider the joint distribution of the true effect size μ_2 and the posterior mean $\hat{\mu}_2^S$, namely,

$$\begin{bmatrix} \mu_2 \\ \hat{\mu}_2^S \end{bmatrix} \mid \hat{\mu}^C \sim \mathcal{N} \left(\begin{bmatrix} \hat{\mu}_2^A \\ \hat{\mu}_2^A \end{bmatrix}, \begin{bmatrix} (\sigma_2^A)^2 & \nu^2 \\ \nu^2 & \nu^2 \end{bmatrix} \right),$$

where $\nu^2 = (\sigma_2^A)^2 - (\sigma_2^S)^2$, and $(\sigma_2^S)^2$ is the posterior variance. Further, consider t_s^* from [Observation 1](#), such that we signal for $\hat{\mu}_2^S \geq t_s^*$ and we do not signal for $\hat{\mu}_2^S < t_s^*$, then the optimal expected cost-to-go (eq. 9) can be written as

$$V_s^* = \int \Phi\left(\frac{t_s^* - \hat{\mu}_2^A}{\nu}\right) dT_{\Pi}^S(x) + \int \Phi\left(\frac{x - \hat{\mu}_2^A}{\sigma_2^A}\right) (-dT_I^S(x)) - \int \tilde{\Phi}\left(\frac{x - \hat{\mu}_2^A}{\sigma_2^A}, \frac{t_s^* - \hat{\mu}_2^A}{\nu}, \frac{\nu}{\sigma_2^A}\right) (dT_{\Pi}^S(x) - dT_I^S(x)). \quad (\text{EC.8})$$

Proof. We note that t_s^* the second period optimal signaling threshold from [Lemma 1](#), such that we signal for $\hat{\mu}_2^S > t_s^*$ and we do not signal for $\hat{\mu}_2^S \leq t_s^*$, then the optimal expected cost-to-go is:

$$\begin{aligned} V_s^* &:= \mathbb{E}_{\mu_2, \hat{\mu}_2^S} [T_I^S(\mu_2) \mathbb{1}[\hat{\mu}_2^S > t_s^*]] + \mathbb{E}_{\mu_2, \hat{\mu}_2^S} [T_{\Pi}^S(\mu_2) \mathbb{1}[\hat{\mu}_2^S \leq t_s^*]] \\ &= \mathbb{E}_{\mu_2, \hat{\mu}_2^S} [T_I^S(\mu_2) \mid \hat{\mu}_2^S > t_s^*] \mathbb{P}(\hat{\mu}_2^S > t_s^*) + \mathbb{E}_{\mu_2, \hat{\mu}_2^S} [T_{\Pi}^S(\mu_2) \mid \hat{\mu}_2^S \leq t_s^*] \mathbb{P}(\hat{\mu}_2^S \leq t_s^*) \quad (\text{Total expectation}) \\ &= \int_{-\infty}^{\infty} T_I^S(x) d\mathbb{P}(\mu_2 \leq x \mid \hat{\mu}_2^S > t_s^*) \mathbb{P}(\hat{\mu}_2^S > t_s^*) + \int_{-\infty}^{\infty} T_{\Pi}^S(x) d\mathbb{P}(\mu_2 \leq x \mid \hat{\mu}_2^S \leq t_s^*) \mathbb{P}(\hat{\mu}_2^S \leq t_s^*) \quad (\text{Riemann-Stieltjes integration}) \\ &= \int_{-\infty}^{\infty} T_I^S(x) d\mathbb{P}(\mu_2 \leq x, \hat{\mu}_2^S > t_s^*) + \int_{-\infty}^{\infty} T_{\Pi}^S(x) d\mathbb{P}(\mu_2 \leq x, \hat{\mu}_2^S \leq t_s^*) \quad (\text{Bayes rule}) \\ &= \lim_{M \rightarrow \infty} - \int_{-M}^M \mathbb{P}(\mu_2 \leq x, \hat{\mu}_2^S > t_s^*) dT_I^S(x) + T_{\Pi}^S(M) \mathbb{P}(\mu_2 \leq M, \hat{\mu}_2^S \leq t_s^*) - \int_{-M}^M \mathbb{P}(\mu_2 \leq x, \hat{\mu}_2^S \leq t_s^*) dT_{\Pi}^S(x) \\ &\quad (\text{Integration by parts \& Definition 1, same ideas as in the proof of Lemma EC.1}) \end{aligned}$$

Noting that $\mathbb{P}(\mu_2 \leq x, \hat{\mu}_2^S > t_s^*) = \mathbb{P}(\mu_2 \leq x) - \mathbb{P}(\mu_2 \leq x, \hat{\mu}_2^S \leq t_s^*)$ and $\mathbb{P}(\mu_2 \leq M, \hat{\mu}_2^S \leq t_s^*) = \mathbb{P}(\hat{\mu}_2^S \leq t_s^*) - \mathbb{P}(\mu_2 > M, \hat{\mu}_2^S \leq t_s^*)$;

$$\begin{aligned} &= \lim_{M \rightarrow \infty} \int_{-M}^M \mathbb{P}(\mu_2 \leq x) (-dT_I^S(x)) + \int_{-M}^M \mathbb{P}(\mu_2 \leq x, \hat{\mu}_2^S \leq t_s^*) dT_I^S(x) \\ &+ \underbrace{\int_{-M}^M \mathbb{P}(\hat{\mu}_2^S \leq t_s^*) dT_{\Pi}^S(x)}_{T_{\Pi}^S(M) = \int_{-M}^M dT_{\Pi}^S(x)} - \underbrace{T_{\Pi}^S(M) \mathbb{P}(\mu_2 > M, \hat{\mu}_2^S \leq t_s^*)}_{\rightarrow 0 (\text{Definition 1})} - \int_{-M}^M \mathbb{P}(\mu_2 \leq x, \hat{\mu}_2^S \leq t_s^*) dT_{\Pi}^S(x) \\ &= \int_{-\infty}^{\infty} \Phi\left(\frac{t_s^* - \hat{\mu}_2^A}{\nu}\right) dT_{\Pi}^S(x) + \int_{-\infty}^{\infty} \Phi\left(\frac{x - \hat{\mu}_2^A}{\sigma_2^A}\right) (-dT_I^S(x)) - \int_{-\infty}^{\infty} \tilde{\Phi}\left(\frac{x - \hat{\mu}_2^A}{\sigma_2^A}, \frac{t_s^* - \hat{\mu}_2^A}{\nu}, \frac{\nu}{\sigma_2^A}\right) (dT_{\Pi}^S(x) - dT_I^S(x)). \end{aligned}$$

In the last expression, we use the bivariate normal distribution [eq. \(EC.1\)](#). $\square$

The second period optimal cost, V_s^* , has the following partial derivatives:

$$\begin{aligned} \frac{dV_s^*}{d\hat{\mu}_2^C} &= \int \frac{-\phi\left(\frac{t_s^* - \hat{\mu}_2^A}{\nu}\right)}{\nu} dT_{\Pi}^S(x) - \int \frac{\phi\left(\frac{x - \hat{\mu}_2^A}{\sigma_2^A}\right)}{\sigma_2^A} (-dT_I^S(x)) \\ &+ \int \left(\frac{\phi\left(\frac{x - \hat{\mu}_2^A}{\sigma_2^A}\right)}{\sigma_2^A} \Phi(\gamma(x)) + \frac{\phi\left(\frac{t_s^* - \hat{\mu}_2^A}{\nu}\right)}{\nu} \Phi\left(\frac{x - t_s^*}{\sigma_2^S}\right) \right) (dT_{\Pi}^S(x) - dT_I^S(x)) \\ &= \int \frac{\phi\left(\frac{x - \hat{\mu}_2^A}{\sigma_2^A}\right)}{\sigma_2^A} \Phi(\gamma(x)) dT_{\Pi}^S(x) + \int \frac{\phi\left(\frac{x - \hat{\mu}_2^A}{\sigma_2^A}\right)}{\sigma_2^A} \left(\Phi(\gamma(x)) - 1 \right) (-dT_I^S(x)), \end{aligned} \quad (\text{EC.9})$$

$$\text{where } \gamma(x) := \frac{(\sigma_2^A)^2 t_s^* - (\sigma_2^S)^2 \hat{\mu}_2^A - \nu^2 x}{\nu \sigma_2^A \sigma_2^S}. \quad (\text{EC.10})$$

$$\frac{dV_s^*}{d\sigma_2^S} = \left(\frac{1}{\nu} \cdot \frac{dt_s^*}{d\sigma_2^S} + \frac{(t_s^* - \hat{\mu}_2^A) \sigma_2^S}{\nu^3} \right) \phi\left(\frac{t_s^* - \hat{\mu}_2^A}{\nu}\right) \underbrace{\left(\int dT_{\Pi}^S(x) - \int \Phi\left(\frac{x - t_s^*}{\sigma_2^S}\right) (dT_{\Pi}^S(x) - dT_I^S(x)) \right)}_{=0 \text{ (Eq. EC.7)}} +$$

$$\begin{aligned}
& \frac{1}{\nu} \phi\left(\frac{\hat{\mu}_2^A - t_S^*}{\nu}\right) \int \phi\left(\frac{x - t_S^*}{\sigma_2^S}\right) (dT_{\text{II}}^S(x) - dT_{\text{I}}^S(x)) \\
&= \frac{1}{\nu} \phi\left(\frac{\hat{\mu}_2^A - t_S^*}{\nu}\right) \int \phi\left(\frac{x - t_S^*}{\sigma_2^S}\right) (dT_{\text{II}}^S(x) - dT_{\text{I}}^S(x)).
\end{aligned} \tag{EC.11}$$

The derivatives are directly obtained from eqs. (EC.2) and (EC.3).

B.2. Bayesian update

Consider the data model introduced in Section 3.1, namely

$$\begin{aligned}
& \text{prior: } \boldsymbol{\mu} \sim \mathcal{N}(\hat{\boldsymbol{\mu}}^A, \boldsymbol{\Sigma}^{bC} + \boldsymbol{\Sigma}^{sC}) \\
& \text{observational data: } \hat{\boldsymbol{\mu}}^O \sim \mathcal{N}(\boldsymbol{\mu} + \boldsymbol{\mu}^{bO}, \boldsymbol{\Sigma}^{bO} + \boldsymbol{\Sigma}^{sO}),
\end{aligned}$$

where $\boldsymbol{\Sigma}^{bC}$ is generalization error, $\boldsymbol{\Sigma}^{sC}$ is sampling error for the clinical trial, $\boldsymbol{\Sigma}^{bO}$ is observational data bias, and $\boldsymbol{\Sigma}^{sO}$ is sampling error of the observational data. The term $\hat{\boldsymbol{\mu}}^A := \hat{\boldsymbol{\mu}}^C - \boldsymbol{\mu}^{bC}$, *i.e.*, the difference between the clinical trial effect size and the generalization bias means, and $\boldsymbol{\mu}^{bO}$ is the observational data bias mean.

We derive the posterior variance using the standard Bayesian update technique. Specifically, given prior $\boldsymbol{\mu} \sim \mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0)$ and data $\mathbf{y} \sim \mathcal{N}(A\boldsymbol{\mu} + \mathbf{b}, \boldsymbol{\Sigma})$, then $\boldsymbol{\mu}|\mathbf{y} \sim \mathcal{N}(\tilde{\boldsymbol{\Sigma}}(A'\boldsymbol{\Sigma}^{-1}(\mathbf{y} - \mathbf{b}) + \boldsymbol{\Sigma}_0^{-1}\boldsymbol{\mu}_0), \tilde{\boldsymbol{\Sigma}})$ for $\tilde{\boldsymbol{\Sigma}} = (\boldsymbol{\Sigma}_0^{-1} + A'\boldsymbol{\Sigma}^{-1}A)^{-1}$. Under our cross-ADR approach, the matrix A corresponds to the identity, then the posterior covariance matrix is obtained as

$$\boldsymbol{\Sigma}^S := \left((\boldsymbol{\Sigma}^{bC} + \boldsymbol{\Sigma}^{sC})^{-1} + (\boldsymbol{\Sigma}^{sO} + \boldsymbol{\Sigma}^{bO})^{-1} \right)^{-1}. \tag{EC.12}$$

For notational simplicity, let us express $\boldsymbol{\Sigma}^{bC} + \boldsymbol{\Sigma}^{sC} = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$, and $\boldsymbol{\Sigma}^{bO} + \boldsymbol{\Sigma}^{sO} = \begin{bmatrix} d & e \\ e & f \end{bmatrix}$. We note that since these are covariance matrices, their determinant is non-negative, that is, $ac - b^2 \geq 0$ and $df - e^2 \geq 0$. Further, the bottom right term of the posterior covariance matrix, $(\sigma_2^S)^2$, is given by

$$(\sigma_2^S)^2 = \frac{c(df - e^2) + f(ac - b^2)}{(a + d)(c + f) - (b + e)^2} = \frac{c|\boldsymbol{\Sigma}^{bO} + \boldsymbol{\Sigma}^{sO}| + f|\boldsymbol{\Sigma}^{bC} + \boldsymbol{\Sigma}^{sC}|}{|\boldsymbol{\Sigma}^{bC} + \boldsymbol{\Sigma}^{sC} + \boldsymbol{\Sigma}^{bO} + \boldsymbol{\Sigma}^{sO}|}. \tag{EC.13}$$

B.2.1. Main paper posterior variances: Under Assumptions 1 and 2, the common ADR is precisely estimated in clinical trials with limited generalization bias¹, so we can simplify the expression in eq. (EC.13) to obtain

$$(\sigma_2^S)^2 = (\sigma_2^A)^2 \left(\frac{(\sigma_2^O)^2 + \kappa}{(\sigma_2^A)^2 + (\sigma_2^O)^2 + \kappa} \right), \tag{EC.14}$$

where we denote $\sigma_2^O := \frac{1}{\sqrt{n^O}}\sigma_2$. The value of $\kappa > 0$ varies depending on the data model and debiasing approach. The value of κ for each model is given by

$$\text{Cross-ADR: } \kappa = (\sigma_2^{bO})^2 (1 - (\rho^{bO})^2), \tag{EC.15}$$

$$\text{Single-ADR: } \kappa = (\sigma_2^{bO})^2. \tag{EC.16}$$

To obtain the posterior variance for the single-ADR data model, we use the adjusted bias covariance matrix $\tilde{\boldsymbol{\Sigma}}^{bO} = \begin{bmatrix} (\sigma_1^{bO})^2 & 0 \\ 0 & (\sigma_2^{bO})^2 \end{bmatrix}$, which is equivalent to assuming $\rho^{bO} = 0$ under Assumptions 1 and 2.

¹ In the terminology of eq. (EC.12), these assumptions yield $a, b \approx 0$, $c = (\sigma_2^A)^2 = \frac{\sigma_2^2}{n^C}$, $d = (\sigma_1^{bO})^2$, $e = \rho^{bO}\sigma_1^{bO}\sigma_2^{bO}$, and $f = (\sigma_2^{bO})^2 + \frac{\sigma_2^2}{n^O}$.

Section C: Proofs

To differentiate the weaker (single-ADR) approach we will use the tilde notation to refer to its parameters, *e.g.*, $\tilde{\sigma}_2^S(>\sigma_2^S)$ denotes the posterior standard deviation of the single-ADR approach. We will also omit the dependency of the cost functions V_A^* and V_S^* on the covariance matrices and use the tilde notation instead whenever the context is clear.

Proof of Lemma 1. The proof is direct from Lemma EC.1 and Observation 1 in Section B. $\square$

Proof of Lemma 2. Recall that at signaling time, $\mu_2|\hat{\mu}^O, \hat{\mu}^C \sim \mathcal{N}(\hat{\mu}_2^S, (\sigma_2^S)^2)$. The expected Type I and II error costs, $\mathbb{E}_{\mu_2|\hat{\mu}^O, \hat{\mu}^C}[T_I^S(\mu_2)]$ and $\mathbb{E}_{\mu_2|\hat{\mu}^O, \hat{\mu}^C}[T_{II}^S(\mu_2)]$, are strictly monotone decreasing and increasing, respectively, in $\hat{\mu}_2^S$; we have defined t_S^* to be the unique value of $\hat{\mu}_2^S$ at which they are equal. For μ^k -loss functions with the same power k , variable substitution $v = \frac{x-t_S^*}{\sigma_2^S}$ yields:

$$\begin{aligned} 0 &= \frac{1}{\sigma_2^S} \left(M_I^S \int_{-\infty}^0 \phi\left(\frac{x-t_S^*}{\sigma_2^S}\right) (-x)^k dx - M_{II}^S \int_0^{\infty} \phi\left(\frac{x-t_S^*}{\sigma_2^S}\right) x^k dx \right) \\ &= (\sigma_2^S)^k \cdot f\left(\frac{t_S^*}{\sigma_2^S}\right), \text{ for} \\ f(z) &:= M_I^S \int_{-\infty}^{-z} \phi(v) (-(v+z))^k dv - M_{II}^S \int_{-z}^{\infty} \phi(v) (v+z)^k dv. \end{aligned}$$

For fixed M_I^S , M_{II}^S , and k , $f(z)$ is a strictly monotone decreasing univariate function, with some unique value z^* such that $f(z^*) = 0$. We conclude that $t_S^* = z^* \sigma_2^S$. $\square$

Proof of Proposition 1. Define the cost difference

$$CD(\hat{\mu}_2^A) := \mathcal{L}_A(\hat{\mu}_2^A, \sigma_2^A, 1) - (\mathcal{L}_A(\hat{\mu}_2^A, \sigma_2^A, 0) + V_S^*(\hat{\mu}^A, \mu^{bO}, \Sigma^{bC}, \Sigma^{sC}, \Sigma^{bO}, \Sigma^{sO}))$$

between label-immediately and wait-and-see. Decompose $T_I^A(\mu_2)$ as $T_I^A(\mu_2) = T_I^S(\mu_2) + \tilde{T}_I(\mu_2)$, noting that $\tilde{T}_I(\mu_2) \geq 0$ (Assumption 3) and $\tilde{T}_I(\mu_2) = 0$ for $\mu_2 > 0$ (Definition 1). Recall that $T_I^A(\mu_2)$ is (weakly) monotone decreasing, however, $\tilde{T}_I(\mu_2)$ is not necessarily monotone decreasing. Then,

$$\begin{aligned} CD(\hat{\mu}_2^A) &= \int \Phi\left(\frac{x-\hat{\mu}_2^A}{\sigma_2^A}\right) (-d\tilde{T}_I(x) + dT_I^S(x)) - \int \Phi^C\left(\frac{x-\hat{\mu}_2^A}{\sigma_2^A}\right) dT_{II}^A(x) - V_S^* \\ &= \int \Phi\left(\frac{x-\hat{\mu}_2^A}{\sigma_2^A}\right) (-d\tilde{T}_I(x)) - \int \Phi^C\left(\frac{x-\hat{\mu}_2^A}{\sigma_2^A}\right) dT_{II}^A(x) + \int \tilde{\Phi}\left(\frac{x-\hat{\mu}_2^A}{\sigma_2^A}, \frac{t_S^*-\hat{\mu}_2^A}{\nu}, \frac{\nu}{\sigma_2^A}\right) (dT_{II}^S(x) - dT_I^S(x)) - \int \Phi\left(\frac{t_S^*-\hat{\mu}_2^A}{\nu}\right) dT_{II}^S(x) \\ &= \underbrace{\left(\int \frac{\phi\left(\frac{x-\hat{\mu}_2^A}{\sigma_2^A}\right)}{\sigma_2^A} (\tilde{T}_I(x) - T_{II}^A(x)) dx \right)}_{L(\hat{\mu}_2^A)} + \underbrace{\left(\int \tilde{\Phi}\left(\frac{x-\hat{\mu}_2^A}{\sigma_2^A}, \frac{t_S^*-\hat{\mu}_2^A}{\nu}, \frac{\nu}{\sigma_2^A}\right) (dT_{II}^S(x) - dT_I^S(x)) - \int \Phi\left(\frac{t_S^*-\hat{\mu}_2^A}{\nu}\right) dT_{II}^S(x) \right)}_{R(\hat{\mu}_2^A)}, \end{aligned}$$

where we use eqs. (EC.5) and (EC.6) from Lemma EC.1, noting that $\tilde{T}_I(x)$ has at most sub-exponential growth (since $T_I^A(x)$ has sub-exponential growth), and the definition in eq. (EC.8) from Lemma EC.2.

Here, $L(\hat{\mu}_2^A)$ represents the expected cost difference between the label-immediately decision and a wait-and-see decision coupled with always signaling in the second period. Meanwhile, $R(\hat{\mu}_2^A)$ represents the expected cost difference between a wait-and-see decision coupled with always signaling in the second period and a wait-and-see decision coupled with optimal signaling in the second period. Note that $L(\hat{\mu}_2^A)$ and $R(\hat{\mu}_2^A)$ take

limiting values $\tilde{T}_1(-\infty) \geq 0$ and $T_1^S(-\infty) > 0$, respectively, as $\hat{\mu}_2^A \rightarrow -\infty$, while they take limiting values $-T_{II}^A(\infty) < 0$ and 0, respectively, as $\hat{\mu}_2^A \rightarrow \infty$. Further, $R(\hat{\mu}_2^A)$ is strictly monotone decreasing in $\hat{\mu}_2^A$:

$$\frac{dR}{d\hat{\mu}_2^A}(\hat{\mu}_2^A) = - \int \frac{\phi\left(\frac{x-\hat{\mu}_2^A}{\sigma_2^A}\right)}{\sigma_2^A} \Phi(\gamma(x)) (dT_{II}^S(x) - dT_I^S(x)) < 0.$$

Note that $L(\hat{\mu}_2^A)$ represents Gaussian smoothing (see [Definition EC.1](#) in [Section A.2](#)) of $\tilde{T}_1(\mu) - T_{II}^A(\mu)$:

$$\begin{aligned} L(\hat{\mu}_2^A) &= \int \left(\tilde{T}_1(\hat{\mu}_2^A + \sigma_2^A z) - T_{II}^A(\hat{\mu}_2^A + \sigma_2^A z) \right) \phi(z) dz, \text{ and} \\ \frac{dL}{d\hat{\mu}_2^A}(\hat{\mu}_2^A) &= \int \left(\tilde{T}_1(\hat{\mu}_2^A + \sigma_2^A z) - T_{II}^A(\hat{\mu}_2^A + \sigma_2^A z) \right) \frac{z\phi(z)}{\sigma_2^A} dz \quad (\text{see Eq. EC.4}). \end{aligned}$$

If $\tilde{T}_1(\mu) = 0$ almost everywhere, then $L(\hat{\mu}_2^A)$ is a strictly decreasing, negative-valued function. Otherwise, $L(\hat{\mu}_2^A)$ is positive for sufficiently small $\hat{\mu}_2^A$ and negative for sufficiently large $\hat{\mu}_2^A$. Because L is a continuous function, there must be some $\hat{\mu}_2^A$ such that $L(\hat{\mu}_2^A) = 0$. At any such $\hat{\mu}_2^A$,

$$\begin{aligned} \frac{dL}{d\hat{\mu}_2^A}(\hat{\mu}_2^A) &= \frac{dL}{d\hat{\mu}_2^A}(\hat{\mu}_2^A) + \frac{\hat{\mu}_2^A}{(\sigma_2^A)^2} L(\hat{\mu}_2^A) = \int \left(\tilde{T}_1(\hat{\mu}_2^A + \sigma_2^A z) - T_{II}^A(\hat{\mu}_2^A + \sigma_2^A z) \right) \frac{(\hat{\mu}_2^A + \sigma_2^A z)\phi(z)}{(\sigma_2^A)^2} dz \\ &= \int \frac{\phi\left(\frac{x-\hat{\mu}_2^A}{\sigma_2^A}\right)}{\sigma_2^A} (\tilde{T}_1(x) - T_{II}^A(x)) \frac{x}{(\sigma_2^A)^2} dx < 0. \end{aligned}$$

The negative derivative follows from the fact that $\tilde{T}_1(\mu) - T_{II}^A(\mu) \leq 0$ for $\mu > 0$ and $\tilde{T}_1(\mu) - T_{II}^A(\mu) \geq 0$ for $\mu < 0$. Since any zero of L has a strictly negative derivative, there is some unique x^* such that $L(x^*) = 0$; $L(\hat{\mu}_2^A)$ is positive for $\hat{\mu}_2^A < x^*$ and negative for $\hat{\mu}_2^A > x^*$.

When $x^* < 0$ and $\hat{\mu}_2^A \in (x^*, 0]$, the same bound establishes that $L(\hat{\mu}_2^A)$ is strictly decreasing:

$$\frac{dL}{d\hat{\mu}_2^A}(\hat{\mu}_2^A) \leq \frac{dL}{d\hat{\mu}_2^A}(\hat{\mu}_2^A) + \frac{\hat{\mu}_2^A}{(\sigma_2^A)^2} L(\hat{\mu}_2^A) < 0.$$

Further, the monotonicity of $T_{II}^A(\mu)$ establishes that $L(\hat{\mu}_2^A)$ is strictly decreasing for $\hat{\mu}_2^A > 0$:

$$\begin{aligned} \frac{dL}{d\hat{\mu}_2^A}(\hat{\mu}_2^A) &= \int \tilde{T}_1(\hat{\mu}_2^A + \sigma_2^A z) \frac{z\phi(z)}{\sigma_2^A} dz - \int \frac{\phi\left(\frac{x-\hat{\mu}_2^A}{\sigma_2^A}\right)}{\sigma_2^A} dT_{II}^A(x) \\ &= \int \frac{\phi\left(\frac{x-\hat{\mu}_2^A}{\sigma_2^A}\right)}{\sigma_2^A} \underbrace{\tilde{T}_1(x) \frac{x - \hat{\mu}_2^A}{(\sigma_2^A)^2}}_{=0 \text{ for } x \geq \hat{\mu}_2^A, \leq 0 \text{ for } x < \hat{\mu}_2^A} dx - \int \frac{\phi\left(\frac{x-\hat{\mu}_2^A}{\sigma_2^A}\right)}{\sigma_2^A} dT_{II}^A(x) < 0. \end{aligned}$$

Define $x^* = -\infty$ for the case where $\tilde{T}_1(\mu) = 0$ almost everywhere. Then we have identified an x^* such that $L(\hat{\mu}_2^A)$ is strictly monotone decreasing for $\hat{\mu}_2^A \in [x^*, \infty)$, decreasing from 0 down to $-T_{II}^A(\infty)$; it takes a positive value for any $\hat{\mu}_2^A < x^*$. Note that x^* is a function of σ_2^A and first-period cost functions $\tilde{T}_1(\mu)$ and $T_{II}^A(\mu)$ but is independent of σ_2^S . Since $R(\hat{\mu}_2^A)$ is also strictly monotone decreasing, we conclude that $CD(\hat{\mu}_2^A)$ is strictly monotone decreasing for $\hat{\mu}_2^A \in [x^*, \infty)$, decreasing from $R(x^*) > 0$ down to $-T_{II}^A(\infty) < 0$. Therefore, there must be some unique threshold $t_A^* > x^*$ such that $CD(\hat{\mu}_2^A) > 0$ for $\hat{\mu}_2^A < t_A^*$ and $CD(\hat{\mu}_2^A) < 0$ for $\hat{\mu}_2^A > t_A^*$.

Lastly, note that $L(\hat{\mu}_2^A)$ is independent of σ_2^S , while $R(\hat{\mu}_2^A)$ is strictly decreasing in σ_2^S :

$$\frac{dR}{d\sigma_2^S} = - \frac{dV_S^*}{d\sigma_2^S} = \frac{-1}{\nu} \phi\left(\frac{\hat{\mu}_2^A - t_A^*}{\nu}\right) \int \phi\left(\frac{x - t_A^*}{\sigma_2^S}\right) (dT_{II}^S(x) - dT_I^S(x)) < 0.$$

This implies that the unique signaling threshold t_A^* at which $CD(t_A^*) = 0$ is strictly decreasing in σ_2^S . $\square$

Referring to $\gamma(x)$ in eq. (EC.10), we state a technical result we use to establish a key comparative static.

LEMMA EC.3 (**Technical result**). $\frac{\tilde{\sigma}_2^S \tilde{\gamma}(\mu)}{\tilde{\nu}} [\Phi(\tilde{\gamma}(\mu)) - \Phi(\gamma(\mu))] + \frac{\tilde{\sigma}_2^S}{\tilde{\nu}} \phi(\tilde{\gamma}(\mu)) - \frac{\sigma_2^S}{\nu} \phi(\gamma(\mu)) > 0$.

Proof: Consider an arbitrary fixed $\hat{\mu}_2^A$; we will prove the desired inequality holds for this $\hat{\mu}_2^A$ and any μ . Defining $h(\mu)$ to be the expression in the statement of the lemma is straightforward, that $h(\mu)$ approaches 0 as $|\mu| \rightarrow \infty$. The derivatives of $h(\mu)$ are

$$\begin{aligned} \frac{dh}{d\mu}(\mu) &= \frac{1}{\sigma_2^A} \left(\eta \phi(\gamma(\mu)) - \Phi(\tilde{\gamma}(\mu)) + \Phi(\gamma(\mu)) \right) \text{ and} \\ \frac{d^2h}{d\mu^2}(\mu) &= \frac{1}{(\sigma_2^A)^2} \left(\frac{\eta\nu}{\sigma_2^S} \phi(\gamma(\mu))\gamma(\mu) + \frac{\tilde{\nu}}{\tilde{\sigma}_2^S} \phi(\tilde{\gamma}(\mu)) - \frac{\nu}{\sigma_2^S} \phi(\gamma(\mu)) \right) \\ &= \frac{\phi(\gamma(\mu))}{(\sigma_2^A)^2} \left(f(\mu) - g(\mu) \right), \text{ for } \eta := \frac{\nu\sigma_2^A}{\sigma_2^S} \left(\frac{\tilde{t}_S^* - \hat{\mu}_2^A}{\tilde{\nu}^2} - \frac{t_S^* - \hat{\mu}_2^A}{\nu^2} \right), \\ f(\mu) &:= \frac{\tilde{\nu}}{\tilde{\sigma}_2^S} \exp \left(\frac{-1}{2} \left((\tilde{\gamma}(\mu))^2 - (\gamma(\mu))^2 \right) \right), \text{ and } g(\mu) := \frac{\nu}{\sigma_2^S} (1 - \eta\gamma(\mu)). \end{aligned}$$

$\tilde{\gamma}(\mu)$ and $\gamma(\mu)$ (defined in eq. EC.10) are both linear, decreasing functions of μ , and $\gamma(\mu)$ has a more strongly negative slope. Therefore, $\frac{-1}{2}((\tilde{\gamma}(\mu))^2 - (\gamma(\mu))^2)$ is a strictly convex quadratic function. This fact, coupled with the fact that $g(\mu)$ is a linear function of μ , easily establishes that $f(\mu) - g(\mu)$ is a strictly convex function that takes positive values as $\mu \rightarrow \infty$ and $\mu \rightarrow -\infty$. A strictly convex function can have at most two zeroes. If $f(\mu)$ and $g(\mu)$ intersect at 0 or 1 points, then $\frac{d^2h}{d\mu^2}(\mu) > 0$ almost everywhere, which is incompatible with the fact that $\frac{dh}{d\mu}(\mu)$ approaches 0 as $|\mu| \rightarrow \infty$. We conclude that $f(\mu)$ and $g(\mu)$ intersect at two points, *i.e.*, $f(\mu) - g(\mu)$ has two zeroes, and that $\frac{d^2h}{d\mu^2}(\mu)$ has a positive-negative-positive pattern for increasing μ .

Given that $\frac{dh}{d\mu}(\mu)$ approaches 0 as $|\mu| \rightarrow \infty$, our pattern of the second derivative implies that $\frac{dh}{d\mu}(\mu)$ has a positive-negative pattern for increasing μ . Finally, given that $h(\mu)$ approaches 0 as $|\mu| \rightarrow \infty$, our pattern of the first derivative implies that $h(\mu) > 0$ for all μ . $\square$

THEOREM EC.1. *The benefit B in eq. (10) is unimodal in $\hat{\mu}_2^A$.*

Proof: Proposition 1 establishes that the stronger (weaker) approach will have some threshold t_A^* ($\tilde{t}_A^*$) such that it will label immediately for $\hat{\mu}_2^A > t_A^*$ ($\tilde{t}_A^*$) and wait and see for $\hat{\mu}_2^A < t_A^*$ ($\tilde{t}_A^*$). Further, this threshold gets smaller as the signaling-time approach gets weaker, meaning $\tilde{t}_A^* < t_A^*$. When $\hat{\mu}_2^A \geq t_A^*$ both approaches label immediately, yielding $B = 0$. When $\tilde{t}_A^* \leq \hat{\mu}_2^A < t_A^*$, then the weaker approach will label immediately while the stronger approach will wait-and-see; Proposition 1 establishes that the benefit (called CD in the proof of Proposition 1) is decreasing in $\hat{\mu}_2^A$ in this region (since $\tilde{t}_A^*, t_A^* > x^*$). Therefore, all that remains to prove is that B is unimodal in $\hat{\mu}_2^A$ is to prove that $B = \tilde{V}_S^* - V_S^*$ is up-down when $\hat{\mu}_2^A < \tilde{t}_A^*$, the region where both approaches wait and see.

Using eq. (EC.9), the derivatives of the benefit in this region are as follows:

$$\begin{aligned} \frac{dB}{d\hat{\mu}_2^A} &= \int \frac{\phi(\frac{\mu - \hat{\mu}_2^A}{\sigma_2^A})}{\sigma_2^A} \left(\Phi(\tilde{\gamma}(\mu)) - \Phi(\gamma(\mu)) \right) (dT_{II}^S(\mu) - dT_I^S(\mu)) \\ \frac{d^2B}{d(\hat{\mu}_2^A)^2} &= \int \frac{\phi(\frac{\mu - \hat{\mu}_2^A}{\sigma_2^A})}{\sigma_2^A} \left[\frac{\mu - \hat{\mu}_2^A}{(\sigma_2^A)^2} \left(\Phi(\tilde{\gamma}(\mu)) - \Phi(\gamma(\mu)) \right) - \left(\frac{\tilde{\sigma}_2^S}{\tilde{\nu}\sigma_2^A} \phi(\tilde{\gamma}(\mu)) - \frac{\sigma_2^S}{\nu\sigma_2^A} \phi(\gamma(\mu)) \right) \right] (dT_{II}^S(\mu) - dT_I^S(\mu)). \end{aligned}$$

Because B is a continuous function of $\hat{\mu}_2^A$ that approaches 0 as $|\hat{\mu}_2^A| \rightarrow \infty$, and because $\frac{dB}{d\hat{\mu}_2^A}$ is continuous, there must be some $\hat{\mu}_2^A$ at which $\frac{dB}{d\hat{\mu}_2^A}(\hat{\mu}_2^A) = 0$. Consider any such $\hat{\mu}_2^A$. Then

$$\begin{aligned} \frac{d^2 B}{d(\hat{\mu}_2^A)^2}(\hat{\mu}_2^A) &= \frac{d^2 B}{d(\hat{\mu}_2^A)^2}(\hat{\mu}_2^A) - \underbrace{\frac{\tilde{t}_A^* - \hat{\mu}_2^A}{\tilde{\nu}^2} \cdot \frac{dB}{d\hat{\mu}_2^A}(\hat{\mu}_2^A)}_{=0} \\ &= - \int \frac{\phi(\frac{\mu - \hat{\mu}_2^A}{\sigma_2^A})}{(\sigma_2^A)^2} \underbrace{\left(\frac{\tilde{\sigma}_2^S \tilde{\gamma}(\mu)}{\tilde{\nu}} [\Phi(\tilde{\gamma}(\mu)) - \Phi(\gamma(\mu))] + \frac{\tilde{\sigma}_2^S}{\tilde{\nu}} \phi(\tilde{\gamma}(\mu)) - \frac{\sigma_2^S}{\nu} \phi(\gamma(\mu)) \right)}_{>0 \text{ (Lemma EC.3)}} (dT_{\text{II}}^S(\mu) - dT_{\text{I}}^S(\mu)) < 0. \end{aligned}$$

Since B has a continuous derivative, the fact that any zero of the derivative has a negative second derivative implies that B is up-down in $\hat{\mu}_2^A$. Note that this result is true for any monotone loss function and independent of Assumptions 1 and 2. $\square$

LEMMA EC.4 (**Technical result**). Consider fixed constants $\hat{\mu}_2^A, z^*, \sigma_2^A > 0, \sigma_2^O := \frac{\sigma_2^S}{\sqrt{n}\sigma} > 0$, and $k \geq 0$, and functions

$$g(\lambda) = (\nu(\lambda))^3 \lambda (\sigma_2^S(\lambda))^{k-1} \phi\left(\frac{\hat{\mu}_2^A - z^* \sigma_2^S(\lambda)}{\nu(\lambda)}\right), \quad \sigma_2^S(\lambda) = \sigma_2^A \sqrt{1 - \frac{(\sigma_2^A)^2}{(\sigma_2^A)^2 + (\sigma_2^O)^2 + \lambda}}, \quad \text{and } \nu(\lambda) = \frac{(\sigma_2^A)^2}{\sqrt{(\sigma_2^A)^2 + (\sigma_2^O)^2 + \lambda}}.$$

Then $g(\lambda)$ is unimodal (up-down) for $\lambda \in [0, \infty)$.

Proof: Since σ_2^S is increasing in λ and ν is decreasing in λ , it is straightforward that terms $\frac{(\sigma_2^S)^2}{\nu^4}, \frac{\sigma_2^S}{\nu^2}, \frac{(\sigma_2^S)^3}{\nu^4}$ and $\frac{1}{\nu^2}$ are increasing in λ . More detailed algebra proves the monotonicity of other functions:

$$\begin{aligned} \frac{d}{d\lambda} \left\{ \frac{(\sigma_2^S)^2}{\nu^4} \left(\frac{2(\sigma_2^A)^4}{\lambda} - 3\nu^2 \right) + (k-1) \right\} &= -(\sigma_2^A)^6 \cdot \frac{2(\sigma_2^O)^2((\sigma_2^A)^2 + (\sigma_2^O)^2) + \lambda^2}{\nu^4 \lambda^2 ((\sigma_2^A)^2 + (\sigma_2^O)^2 + \lambda)^2} < 0, \\ \frac{d}{d\lambda} \left\{ \frac{2(\sigma_2^A)^4}{\lambda \nu^2} - 3 + \frac{(k-1)\nu^2}{(\sigma_2^S)^2} \right\} &\leq \frac{d}{d\lambda} \left\{ \frac{2(\sigma_2^A)^4}{\lambda \nu^2} - 3 - \frac{\nu^2}{(\sigma_2^S)^2} \right\} \\ &= -(\sigma_2^A)^4 \cdot \frac{c_0 + c_1 \lambda + c_2 \lambda^2 + c_3 \lambda^3}{\lambda^2 (\sigma_2^S)^4 ((\sigma_2^A)^2 + (\sigma_2^O)^2 + \lambda)^3} \text{ for } c_0, c_1, c_2, c_3 > 0 \\ &< 0, \text{ and} \\ \frac{d}{d\lambda} \left\{ \frac{1}{\sigma_2^S} + 2 \frac{\sigma_2^S - \sigma_2^A}{\nu^2} \right\} &= \frac{-(\sigma_2^A)^3 - 2(\sigma_2^S)^3 + 3\sigma_2^A (\sigma_2^S)^2}{2(\sigma_2^A)^3 (\sigma_2^S)^3} < 0. \end{aligned}$$

The bound used in the second expression follows from the fact that $\frac{\nu^2}{(\sigma_2^S)^2}$ is decreasing in λ and $k \geq 0$, and the monotonicity follows from careful algebra.² The conclusion of monotonicity in the third case follows from the fact that numerator $-(\sigma_2^A)^3 - 2(\sigma_2^S)^3 + 3\sigma_2^A (\sigma_2^S)^2$ is strictly increasing in σ_2^S , meaning it is upper bounded by 0 (its value when $\sigma_2^S = \sigma_2^A$).

Proving the result follows by rearranging the terms of $\frac{dg}{d\lambda}(\lambda)$. When $\hat{\mu}_2^C z^* \leq 0$, then

$$\frac{dg}{d\lambda}(\lambda) = \frac{\phi(\frac{\hat{\mu}_2^A - z^* \sigma_2^S}{\nu}) \nu^7 \lambda (\sigma_2^S)^{k-3}}{2(\sigma_2^A)^4} \left(\underbrace{\frac{(\sigma_2^S)^2}{\nu^4} \left(\frac{2(\sigma_2^A)^4}{\lambda} - 3\nu^2 \right) + (k-1)}_{\downarrow \text{ in } \lambda} - \underbrace{\frac{(\sigma_2^S)^2}{\nu^4} ((\hat{\mu}_2^A)^2 + (z^*)^2 (\sigma_2^A)^2)}_{\uparrow \text{ in } \lambda} + \underbrace{\left(\frac{\sigma_2^S}{\nu^2} + \frac{2(\sigma_2^S)^3}{\nu^4} \right)}_{\uparrow \text{ in } \lambda} \underbrace{\hat{\mu}_2^A z^*}_{\leq 0} \right).$$

² $c_0 = 2((\sigma_2^A)^2 + (\sigma_2^O)^2)^2 (\sigma_2^O)^4$, $c_1 = 2(\sigma_2^O)^2 ((\sigma_2^A)^2 + (\sigma_2^O)^2) (2(\sigma_2^A)^2 + 3(\sigma_2^O)^2)$, $c_2 = (\sigma_2^A)^4 + 7(\sigma_2^A)^2 (\sigma_2^O)^2 + 6(\sigma_2^O)^4$, and $c_3 = (\sigma_2^A)^2 + 2(\sigma_2^O)^2$.

Meanwhile, when $\hat{\mu}_2^C z^* \geq 0$, then

$$\frac{dg}{d\lambda}(\lambda) = \frac{\phi(\frac{\hat{\mu}_2^A - z^* \sigma_2^S}{\nu}) \nu^5 \lambda (\sigma_2^S)^{k-1}}{2(\sigma_2^A)^4} \left(\underbrace{\frac{2(\sigma_2^A)^4}{\lambda \nu^2} - 3 + \frac{(k-1)\nu^2}{(\sigma_2^S)^2}}_{\downarrow \text{ in } \lambda} - \underbrace{\frac{1}{\nu^2}}_{\uparrow \text{ in } \lambda} \underbrace{(\hat{\mu}_2^A - z^* \sigma_2^A)^2}_{\geq 0} + \underbrace{\left(\frac{1}{\sigma_2^S} + 2\frac{\sigma_2^S - \sigma_2^A}{\nu^2}\right)}_{\downarrow \text{ in } \lambda} \underbrace{\hat{\mu}_2^A z^*}_{\geq 0} \right).$$

In both cases, $\frac{dg}{d\lambda}(\lambda)$ is the product of a positive term times a term that decreases in λ . Since $g(0) = 0$ and $\lim_{\lambda \rightarrow \infty} g(\lambda) = 0$, $g(\lambda)$ must have an up-down shape. $\square$

THEOREM EC.2. *Under Assumptions 1 and 2, the benefit B in eq. (10) is unimodal up-down in σ_2^{bO} for the μ^k -loss function.*

Proof: Define σ_2^S and $\tilde{\sigma}_2^S$ to be the posterior standard deviations for our approach and the single-ADR approach. Note that both are strictly increasing functions of σ_2^{bO} . We can rewrite eq. (EC.14) as follows³:

$$(\tilde{\sigma}_2^S)^2 = (\sigma_2^A)^2 \left(1 - \frac{(\sigma_2^A)^2}{(\sigma_2^A)^2 + (\sigma_2^O)^2 + (\sigma_2^{bO})^2} \right), \text{ and } (\sigma_2^S)^2 = (\sigma_2^A)^2 \left(1 - \frac{(\sigma_2^A)^2}{(\sigma_2^A)^2 + (\sigma_2^O)^2 + \delta(\sigma_2^{bO})^2} \right).$$

Since V_S^* , the expected cost of the signaling-time decision, is strictly monotone increasing in σ_2^S (Eq. EC.11), the expected cost for both our approach and the single-ADR approach are also strictly monotone increasing in σ_2^{bO} . Meanwhile, the expected cost of labeling in the first period is not a function of σ_2^{bO} . Thus, there must be some $0 \leq t_1 \leq t_2 \leq \infty$ such that both approaches wait-and-see for $\sigma_2^{bO} \in [0, t_1)$, the single-ADR approach labels immediately while our approach waits-and-sees for $\sigma_2^{bO} \in [t_1, t_2)$, and both approaches label immediately for $\sigma_2^{bO} \geq t_2$. B is (weakly) monotone decreasing for $\sigma_2^{bO} \geq t_1$, so all that remains to prove the result is to show that B is up-down in the regime where both approaches choose to wait-and-see.

In this regime, and using eq. (EC.11), the derivative of the benefit with respect to σ_2^{bO} can be written as

$$\frac{dB}{d\sigma_2^{bO}} = \frac{d\tilde{\sigma}_2^S}{d\sigma_2^{bO}} \frac{dV_S^*}{d\sigma_2^S}(\tilde{\sigma}_2^S) - \frac{d\sigma_2^S}{d\sigma_2^{bO}} \frac{dV_S^*}{d\sigma_2^S}(\sigma_2^S) = \frac{1}{(\sigma_2^A)^4 \sigma_2^{bO}} \left(\frac{\tilde{\nu}^4 (\sigma_2^{bO})^2}{\tilde{\sigma}_2^S} \frac{dV_S^*}{d\sigma_2^S}(\tilde{\sigma}_2^S) - \frac{\nu^4 \delta (\sigma_2^{bO})^2}{\sigma_2^S} \frac{dV_S^*}{d\sigma_2^S}(\sigma_2^S) \right).$$

For the μ^k -loss function the benefit eq. (10) can be simplified using the form of the loss function from Definition 2 and the fact that the signaling threshold t_S^* takes the form $t_S^* = z^* \sigma_2^S$ for a constant z^* (Lemma 2):

$$\frac{dV_S^*}{d\sigma_2^S}(\sigma_2^S) = \frac{c(\sigma_2^S)^k}{\nu} \phi\left(\frac{\hat{\mu}_2^A - z^* \sigma_2^S}{\nu}\right), \text{ for } c := \begin{cases} (M_I^S + M_{II}^S) \phi(z^*) & \text{if } k = 0, \text{ and} \\ k M_I^S \int_{-\infty}^{-z^*} \phi(v) (-v + z^*)^{k-1} dv + k M_{II}^S \int_{-z^*}^{\infty} \phi(v) (v + z^*)^{k-1} dv & \text{if } k > 0. \end{cases} \quad (\text{EC.17})$$

Therefore, the derivative of the benefit with respect to σ_2^{bO} can be re-expressed as

$$\frac{dB}{d\sigma_2^{bO}} = \underbrace{\frac{c}{(\sigma_2^A)^4 \sigma_2^{bO}}}_{>0} \left(g((\sigma_2^{bO})^2) - g(\delta(\sigma_2^{bO})^2) \right),$$

for $g(\lambda)$ as defined in Lemma EC.4.

By Lemma EC.4, there exists some λ^* such that $g(\lambda)$ is increasing for $\lambda < \lambda^*$ and decreasing for $\lambda > \lambda^*$. When $(\sigma_2^{bO})^2 \leq \lambda^*$, then $g((\sigma_2^{bO})^2) \geq g(\delta(\sigma_2^{bO})^2)$, so $\frac{dB}{d\sigma_2^{bO}}(\sigma_2^{bO}) \geq 0$, and similarly $\frac{dB}{d\sigma_2^{bO}}(\sigma_2^{bO}) \leq 0$ when $\delta(\sigma_2^{bO})^2 \geq \lambda^*$. When $\delta(\sigma_2^{bO})^2 < \lambda^* < (\sigma_2^{bO})^2$, then $g((\sigma_2^{bO})^2)$ is decreasing in σ_2^{bO} while $g(\delta(\sigma_2^{bO})^2)$ is increasing in σ_2^{bO} , implying that $\frac{dB}{d\sigma_2^{bO}}(\sigma_2^{bO})$ is positive-negative in this region. $\square$

³ We borrow notation $0 < \delta := 1 - (\rho^{bO})^2$ to indicate the strength of debiasing under our approach; smaller δ indicate stronger debiasing.

THEOREM EC.3. *Under Assumptions 1 and 2, and under the μ^k -loss function, B is monotone increasing in n^O whenever $k \leq 1$ and $\hat{\mu}_2^A z^* \leq 0$. B is bimodal (up-down) in n^O whenever $k > 1$, and B has indeterminate behavior in n^O when $k \leq 1$ and $\hat{\mu}_2^A z^* > 0$.*

Proof: V_S^* and $\tilde{V}_S^*$, the expected costs of the signaling-time decision for our approach and the single-ADR approach, are strictly monotone increasing in σ_2^S and $\tilde{\sigma}_2^S$, respectively. Since both σ_2^S and $\tilde{\sigma}_2^S$ decrease in n^O , the expected cost of the wait-and-see decision for both our approach and the single-ADR approach are also strictly monotone decreasing in n^O . Meanwhile, the expected cost of labeling in the first period is not a function of n^O . Thus, there must be some $0 \leq t_1 \leq t_2 \leq \infty$ such that both approaches label immediately for $n^O \in [0, t_1)$, the single-ADR approach labels immediately while our approach waits-and-sees for $n^O \in [t_1, t_2)$, and both approaches wait-and-see for $n^O \geq t_2$. B is (weakly) monotone increasing for $n^O < t_2$, so all that remains to prove the result is to show that B is increasing in the regime where both approaches choose to wait-and-see.

In this regime, we can follow the development from the [Theorem EC.2](#) to write the derivative of the benefit with respect to n^O as

$$\begin{aligned} \frac{dB}{dn^O} &= \frac{d\tilde{\sigma}_2^S}{dn^O} \frac{dV_S^*}{d\sigma_2^S}(\tilde{\sigma}_2^S) - \frac{d\sigma_2^S}{dn^O} \frac{dV_S^*}{d\sigma_2^S}(\sigma_2^S) \\ &= \frac{\sigma_2^S}{2(\sigma_2^A)^4(n^O)^2} \left(\frac{\nu^4}{\sigma_2^S} \frac{dV_S^*}{d\sigma_2^S}(\sigma_2^S) - \frac{\tilde{\nu}^4}{\tilde{\sigma}_2^S} \frac{dV_S^*}{d\sigma_2^S}(\tilde{\sigma}_2^S) \right) = \frac{c\sigma_2^S}{2(\sigma_2^A)^4(n^O)^2} \left(g(\sigma_2^S) - g(\tilde{\sigma}_2^S) \right), \text{ for} \\ g(\sigma_2^S) &:= (\sigma_2^S)^{k-1} \nu^3 \phi \left(\frac{\hat{\mu}_2^A - z^* \sigma_2^S}{\nu} \right) \text{ and constant } c > 0 \text{ from eq. (EC.17).} \end{aligned}$$

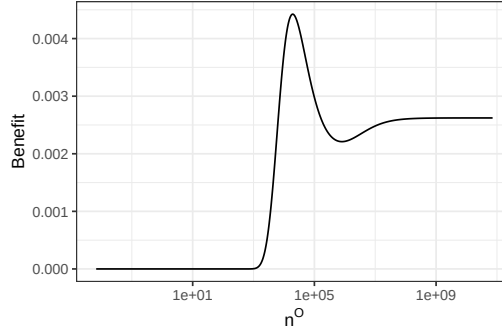
Case: $k > 1$. To analyze this case, we argue that $g(\sigma_2^S)$ is an up-down function, which will lead us to conclude that the benefit is up-down in n^O . To make this argument, we note that

$$\begin{aligned} \frac{dg(\sigma_2^S)}{d\sigma_2^S} &= \frac{(\sigma_2^S)^{k-2}}{\nu} \phi \left(\frac{\hat{\mu}_2^A - z^* \sigma_2^S}{\nu} \right) \left((k-1)(\sigma_2^A)^4 + \hat{\mu}_2^A z^* (\sigma_2^A)^2 \sigma_2^S - ((\sigma_2^A)^2(2k+1+(z^*)^2) + (\hat{\mu}_2^A)^2)(\sigma_2^S)^2 \right. \\ &\quad \left. + \hat{\mu}_2^A z^* (\sigma_2^S)^3 + (k+2)(\sigma_2^S)^4 \right). \end{aligned}$$

We have expressed $g'(\sigma_2^S)$ as a positive term times a term that is quartic in $\sigma_2^S \in [0, \sigma_2^A]$. From $k > 1$, the quartic term takes positive value at $\sigma_2^S = 0$ and as $\sigma_2^S \rightarrow \infty$ or $\sigma_2^S \rightarrow -\infty$. Further, the quartic term evaluates to value $-(\sigma_2^A)^2(\hat{\mu}_2^A - z^* \sigma_2^A)^2 \leq 0$ at $\sigma_2^S = \sigma_2^A$ and evaluates to value $-(\sigma_2^A)^2(\hat{\mu}_2^A + z^* \sigma_2^A)^2 \leq 0$ at $\sigma_2^S = -\sigma_2^A$. As a result, the quartic term must have one root in $(\infty, -\sigma_2^A]$, one root in $[-\sigma_2^A, 0)$, one root in $(0, \sigma_2^A]$, and one root in $[\sigma_2^A, \infty)$. Hence, we conclude there is some $\sigma^* \in [0, \sigma_2^A]$ such that $g'(\sigma) > 0$ for $\sigma < \sigma^*$ and $g'(\sigma) < 0$ for $\sigma > \sigma^*$. From this, we conclude that $\frac{dB}{dn^O} < 0$ when $\sigma_2^S < \tilde{\sigma}_2^S \leq \sigma^*$ (when n^O is very large) and that $\frac{dB}{dn^O} > 0$ when $\sigma^* \leq \sigma_2^S < \tilde{\sigma}_2^S$ (when n^O is very small). When $\sigma_2^S < \sigma^* < \tilde{\sigma}_2^S$, $g(\sigma_2^S)$ increases in σ_2^S while $g(\tilde{\sigma}_2^S)$ decreases in $\tilde{\sigma}_2^S$, meaning that $\frac{dB}{dn^O}$ has a positive-negative pattern in n^O .

Case: $k \leq 1$ and $\hat{\mu}_2^A z^* \leq 0$. Clearly $\nu = \sqrt{(\sigma_2^A)^2 - (\sigma_2^S)^2}$ is decreasing in σ_2^S . From $k \leq 1$, it is also clear that $(\sigma_2^S)^{k-1}$ is decreasing in σ_2^S . From $z^* \hat{\mu}_2^A \leq 0$, we conclude that $|\frac{\hat{\mu}_2^A - z^* \sigma_2^S}{\nu}|$ is increasing in σ_2^S , meaning that $\phi(\frac{\hat{\mu}_2^A - z^* \sigma_2^S}{\nu})$ is decreasing in σ_2^S . Putting these results together, $g(\sigma_2^S)$ is a strictly decreasing function under this case. Then, since $\sigma_2^S < \tilde{\sigma}_2^S$, we obtain that the benefit of our approach is strictly increasing in n^O .

Case: $k \leq 1$ and $\hat{\mu}_2^A z^* > 0$. We provide an example of indeterminate behavior in [Figure EC.1](#). $\square$

Figure EC.1 Example of up-down-up benefit pattern in n^O for the μ^k -loss function with $k = 0$ 

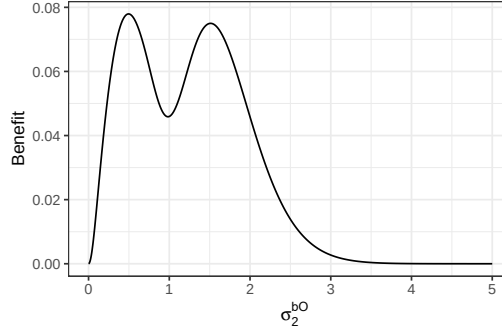
Note. The plot shows the second-period cost difference $\tilde{V}_S^* - V_S^*$ (this is the relevant portion of the benefit whenever $M_{II}^A \approx 0$). The parameters $\hat{\mu}_1^A$, μ_1^{bO} , and σ_1^{bO} do not affect the benefit (see [Theorem 1](#)), and under Assumptions [1](#) and [2](#) the parameters ρ^{sC} , ρ^{sO} , and ρ^{bC} also do not affect the benefit. We only specify the remaining parameters, namely, signaling-time cost function parameters $M_I^S = 1$, $M_{II}^S = 133,000$, and $k = 0$, as well as system parameters $\sigma_2 = 100$, $n^C = 6,000$, $\mu_2^{bO} = 0$, $\sigma_2^{bO} = 0.1$, $\rho^{bO} = 0.95$, $\sigma_2^{bC} = 0$, and $\hat{\mu}_2^A = -3.75$.

Proof of Theorem 1. We note that the single-ADR model posterior variance is larger, *i.e.*, $\tilde{\sigma}_2^S > \sigma_2^S$. Hence, $\tilde{\nu} < \nu$. We next show each parameter's comparative static for either the monotone loss and/or the power loss function and provide counterexamples to justify why the behavior is indeterminate in some cases.

- The benefit is increasing in ρ^{bO} for the general class of monotone loss functions. We note that σ_2^S in [eq. \(EC.15\)](#) is decreasing in $|\rho^{bO}|$, while $\tilde{\sigma}_2^S$ is independent of $|\rho^{bO}|$. Since V_S^* is strictly monotone increasing in σ_2^S and $\tilde{\sigma}_2^S$ ([eq. EC.11](#)), the expected cost of the wait-and-see decision is strictly monotone decreasing in $|\rho^{bO}|$ for our approach and constant for the single-ADR approach. Meanwhile, the expected cost of labeling in the first period is not a function of $|\rho^{bO}|$. We conclude that our approach's expected cost is (weakly) decreasing in $|\rho^{bO}|$ while the single-ADR approach's expected cost is constant in $|\rho^{bO}|$, yielding a (weakly) increasing benefit in $|\rho^{bO}|$.
- The benefit being constant in μ_1^{bO} , σ_1^{bO} , and $\hat{\mu}_1^A$ is direct from noting that the posterior variances $(\sigma_2^S)^2$ and $(\tilde{\sigma}_2^S)^2$ in [eqs. \(EC.15\)](#) and [\(EC.16\)](#) do not depend on μ_1^{bO} , σ_1^{bO} or $\hat{\mu}_1^A$ so none of the expected costs depend on either parameter. Note that this result is true for the general class of monotone loss functions.
- We subtract out the bias mean before performing debiasing, so μ_2^{bO} has no effect on expected loss when all parameters are known.
- The benefit is unimodal (up-down) with respect to $\hat{\mu}_2^A$, as shown in [Theorem EC.1](#). We note that unimodal behavior is true for the general class of monotone loss functions.
- The unimodal behavior of the benefit with respect to σ_2^{bO} for the μ^k -loss function is presented in [Theorem EC.2](#). Note that this behavior is not necessarily true for the class of general monotone loss functions. To illustrate this, let us consider the following monotone loss function:

$$T_I^S(\mu) = \begin{cases} 300 & \text{if } \mu \leq -5 \\ 1 & \text{if } -5 < \mu < 0 \end{cases}$$

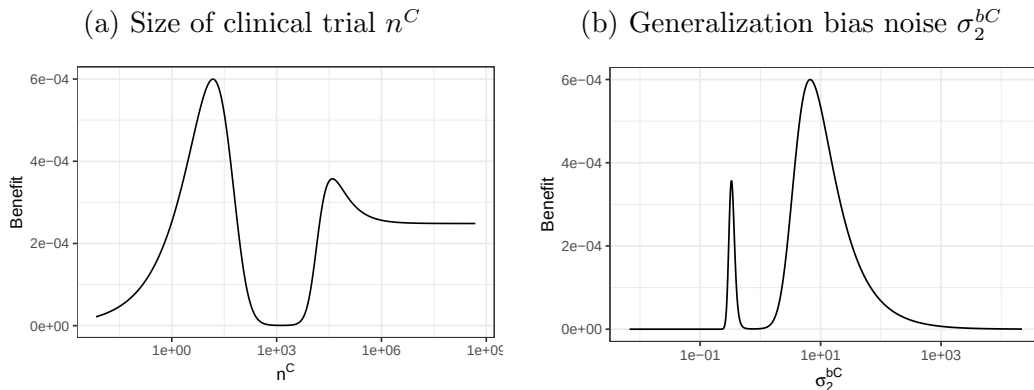
$$T_{II}^S(\mu) = 3000 \text{ if } \mu > 0.$$

Figure EC.2 Example of bimodal benefit pattern in σ_2^{bO} for a general monotonic loss function

Note. The plot shows the second-period cost difference $\tilde{V}_S^* - V_S^*$ (this is the relevant portion of the benefit whenever $M_{II}^A \approx 0$). The parameters $\hat{\mu}_1^A$, μ_1^{bO} , and σ_1^{bO} do not affect the benefit (see [Theorem 1](#)), and under Assumptions 1 and 2 the parameters ρ^{sC} , ρ^{sO} , and ρ^{bC} also do not affect the benefit. We only specify the remaining parameters, namely, $\sigma_2 = 15$, $n^C = 100$, $n^O = 22,500$, $\mu_2^{bO} = 0$, $\rho^{bO} = 1/\sqrt{2}$, $\sigma_2^{bC} = 0$, and $\hat{\mu}_2^A = 0$.

Then, [Figure EC.2](#) shows a bimodal pattern when both approaches choose to wait-and-see. Hence, we conclude that, for the general class of monotone loss functions, the benefit varies in an indeterminate way with σ_2^{bO} .

- Indeterminate behavior for σ_2 , n^C , and σ_2^{bC} . We note that under Assumptions 1 and 2, the parameters σ_2 , n^C , and σ_2^{bC} impact the benefit through the prior standard deviation σ_2^A as well as posterior standard deviations σ_2^S and $\tilde{\sigma}_2^S$. We provide examples of indeterminate behavior in [Figure EC.3](#).
- The benefit is monotone increasing in n^O for the μ^k -loss functions whenever $k \leq 1$ and $\hat{\mu}_2^A z^* \leq 0$ unimodal (up-down) in n^O whenever $k > 1$, and is indeterminate otherwise. This result is presented in [Theorem EC.3](#). $\square$

Figure EC.3 Examples of indeterminate benefit patterns for μ^k -loss function with $k = 0$.

Note: The plots show the second-period cost difference $\tilde{V}_S^* - V_S^*$ (this is the relevant portion of the benefit whenever $M_{II}^A \approx 0$). The parameters $\hat{\mu}_1^A$, μ_1^{bO} , and σ_1^{bO} do not affect the benefit (see [Theorem 1](#)), and under Assumptions 1 and 2, ρ^{sC} , ρ^{sO} , and ρ^{bC} also do not affect the benefit. We only specify the remaining parameters: $M_I^S = 1$, $M_{II}^S = 0.008$, $k = 0$, $\sigma_2 = 26$, $n^C = 10,000$ (right), $n^O = 500$, $\mu_2^{bO} = 0$, $\sigma_2^{bO} = 3$, $\rho^{bO} = 0.7$, $\sigma_2^{bC} = 0.4$ (left), and $\hat{\mu}_2^A = 1$.

Section D: Presence of generalization bias in common ADR

We relax [Assumption 2](#), to consider the case $\sigma_1^{bC} \geq 0$. The results in [Lemma 2](#) and [Proposition 1](#) remain unchanged. We need to show how the comparative statics results in [Theorem 1](#) change due to the presence of generalization bias in the common ADR.

Note that relaxing [Assumption 2](#) results in the following covariance matrices:

$$\Sigma^{bC} + \Sigma^{sC} = \begin{bmatrix} (\sigma_1^{bC})^2 & \rho^{bC} \sigma_1^{bC} \sigma_2^{bC} \\ \rho^{bC} \sigma_1^{bC} \sigma_2^{bC} & (\sigma_2^{bC})^2 \end{bmatrix} := \begin{bmatrix} a & b \\ b & c \end{bmatrix}, \text{ and } \Sigma^{bO} + \Sigma^{sO} = \begin{bmatrix} (\sigma_1^{bO})^2 & \rho^{bO} \sigma_1^{bO} \sigma_2^{bO} \\ \rho^{bO} \sigma_1^{bO} \sigma_2^{bO} & (\sigma_2^{bO})^2 + (\sigma_2^O)^2 \end{bmatrix} := \begin{bmatrix} d & e \\ e & f \end{bmatrix},$$

where for notation convenience we define $(\sigma_2^A)^2 := (\sigma_2^{bC})^2 + \frac{1}{n^C}(\sigma_2)^2$ and $(\sigma_2^O)^2 := \frac{1}{n^O}(\sigma_2)^2$. Further, we note that the single-ADR approach ignores any links between the common and rare ADR. In our model, this is equivalent to relying on adjusted matrices $\tilde{\Sigma}^{bC} = \begin{bmatrix} (\sigma_1^{bC})^2 & 0 \\ 0 & (\sigma_2^{bC})^2 \end{bmatrix}$ and $\tilde{\Sigma}^{bO} = \begin{bmatrix} (\sigma_1^{bO})^2 & 0 \\ 0 & (\sigma_2^{bO})^2 \end{bmatrix}$ in the Bayesian update step of the single-ADR approach.

THEOREM EC.4. *Under [Assumption 1](#), the benefit*

$$B := V_A^*(\hat{\mu}^A, \mu^{bO}, \tilde{\Sigma}^{bC}, \Sigma^{sC}, \tilde{\Sigma}^{bO}, \Sigma^{sO}) - V_A^*(\hat{\mu}^A, \mu^{bO}, \Sigma^{bC}, \Sigma^{sC}, \Sigma^{bO}, \Sigma^{sO})$$

is non-negative and satisfies the comparative statics in [Table EC.1](#).

Table EC.1 Comparative statics of the benefit under $\sigma_1^{bC} \geq 0$ (i.e., relaxing [Assumption 2](#))

Parameter		Monotone Loss	μ^k -Loss (Power Loss)
Clinical trial	$\hat{\mu}_1^A$	constant	constant
	$\hat{\mu}_2^A$	unimodal	unimodal
	σ_2	indeterminate	indeterminate
	n^C	indeterminate	indeterminate
	(*) ρ^{bC}	down-up	down-up
	(*) σ_1^{bC}	up-down or down-up	up-down or down-up
	σ_2^{bC}	indeterminate	indeterminate
Observational	μ_i^{bO}	constant	constant
	(*) ρ^{bO}	down-up	down-up
	(*) σ_1^{bO}	up-down or down-up	up-down or down-up
	(*) σ_2^{bO}	indeterminate	indeterminate
	(*) n^O	indeterminate	indeterminate

Note: Benefit is, under [Assumption 1](#), independent of parameters ρ^{sC} and ρ^{sO} . (*) shows that the parameter has either an updated benefit pattern or is a new comparative statics relative to [Theorem 1](#).

Proof: The parameters in [Theorem 1](#) with indeterminate benefit pattern remain the same, so we do not repeat the counterexamples here. For $\hat{\mu}_1^A$ and $\hat{\mu}_2^A$ the proof of [Theorem 1](#) is still valid. In the proof, we focus on the parameters with new insights, those marked with (*) in [Table EC.1](#).

- The benefit is down-up in ρ^{bO} and ρ^{bC} . We note that both correlations affect the benefit only through the cross-ADR approach posterior variance σ_2^S (the single-ADR posterior variance does not depend on these correlations). We look at the partial with respect to $e := \rho^{bO} \sigma_1^{bO} \sigma_2^{bO}$, namely,

$$\frac{\partial(\sigma_2^S)^2}{\partial e} = 2 \cdot \frac{bf(ac - b^2 + cd) - (c(ac - b^2) + c^2d + b^2f)e + bce^2}{((a + d)(c + f) - (b + e)^2)^2}.$$

The denominator is positive, so the comparative statistics are given by the sign of the numerator. We notice that this is a quadratic function of e , so we are signing a quadratic function over its range $e \in [-\sqrt{df}, \sqrt{df}]$. Then the derivative of the numerator with respect to e is

$$2bce - (c(ac - b^2) + c^2d + b^2f) \leq \underbrace{2|b|c\sqrt{df}}_{:=A \geq 0} - \underbrace{(c(ac - b^2) + c^2d + b^2f)}_{:=B \geq 0}.$$

The bound here comes from noting that the first term is maximized at the largest value of the interval, $e = \sqrt{df}$, when $b \geq 0$ and at the bottom of the interval, $e = -\sqrt{df}$, when $b \leq 0$. We can sign this by noting that

$$\begin{aligned} B^2 - A^2 &= c^2(ac - b^2)^2 + c^4d^2 + b^4f^2 + 2c^3d(ac - b^2) + 2b^2cf(ac - b^2) - 2b^2c^2df \\ &\geq c^4d^2 + b^4f^2 - 2b^2c^2df = (c^2d - b^2f)^2 \geq 0. \end{aligned}$$

We conclude that the numerator of $\frac{\partial(\sigma_2^S)^2}{\partial e}$ is decreasing for $e \in [-\sqrt{df}, \sqrt{df}]$, which means that the posterior variance has a positive-negative pattern. Since cost V_S^* is strictly monotone increasing in σ_2^S , the expected signaling cost is up-down in ρ^{bO} . To conclude that the benefit is down-up in ρ^{bO} , we compare the costs of both approaches. The expected cost of labeling in the first period $c_1 := \mathcal{L}_A(\hat{\mu}_2^A, \sigma_2^A, 1)$ and not labeling in the first period $c_2 := \mathcal{L}_A(\hat{\mu}_2^A, \sigma_2^A, 0)$ are the same for both, the single-ADR and the cross-ADR debiasing approaches. The optimal expected signaling costs for the single-ADR approach is $c_3 := \tilde{V}_S^*$, which is independent of ρ^{bO} . The optimal signaling cost for our approach is $c(\rho^{bO}) := V_S^*(\rho^{bO}) < c_3$ and up-down in ρ^{bO} from above. Thus, the total cost of single-ADR is $\min\{c_1, c_2 + c_3\}$ and for our approach is $\min\{c_1, c_2 + c(\rho^{bO})\}$. From this, we note that our approach can result in optimal decisions: wait-and-see | label-immediately | wait-and-see, as ρ^{bO} increases from -1 to +1. We further notice that the label-immediately mid region may or may not exist (*e.g.*, it does not exist whenever $c_1 \geq c_2 + c_3$). Then, the benefit $\min\{c_1, c_2 + c_3\} - \min\{c_1, c_2 + c(\rho^{bO})\}$ is either down-zero-up (if the label-immediately region exists) or down-up in ρ^{bO} .

By the symmetry of the covariance matrices used in the Bayesian update, it is clear that we will obtain identical comparative statics for ρ^{bC} .

- The benefit can be up-down or down-up in σ_1^{bO} and σ_1^{bC} . Note that the common ADR bias noise only affects the benefit through the posterior variance $(\sigma_2^S)^2$ of the cross-ADR approach. Note that

$$\frac{\partial(\sigma_2^S)^2}{\partial \sigma_1^{bO}} = 2 \cdot \frac{d_1 + d_2\sigma_1^{bO} + d_3(\sigma_1^{bO})^2}{((a+d)(c+f) - (b+e)^2)^2},$$

with $d_1 := \rho^{bO}\sigma_2^{bO}(ac - b^2)bf$, $d_2 := (-ac^2 + b^2c - b^2f)(\rho^{bO})^2(\sigma_2^{bO})^2 + b^2f^2$, and $d_3 := bc\rho^{bO}\sigma_2^{bO}((\rho^{bO})^2(\sigma_2^{bO})^2 - f)$. The derivative has a positive denominator and a numerator that is quadratic in σ_1^{bO} , so we can have at most two sign changes in the comparative statics. The zeros of the numerator occur at $\sigma_1^{bO} = \frac{-d_2 \pm \sqrt{d_2^2 - 4d_1d_3}}{2d_3}$. Noting that

$$4d_1d_3 = \underbrace{4b^2c(\rho^{bO})^2(\sigma_2^{bO})^2f}_{\geq 0} \underbrace{(ac - b^2)}_{\geq 0} \underbrace{(((\rho^{bO})^2 - 1)(\sigma_2^{bO})^2 - (\sigma_2^2/n^O))}_{\geq 0} \geq 0.$$

Thus, the numerator quadratic function has at most one zero at some $\sigma_1^{bO} > 0$, which implies that the posterior variance is either up-down or down-up in σ_1^{bO} . Since the optimal expected signaling cost (V_S^*) is increasing in σ_2^S , we conclude the optimal signaling cost is also either up-down or down-up in σ_1^{bO} .

To conclude that the benefit is down-up in σ_1^{bO} , we compare the costs of both approaches. The expected cost of labeling $c_1 := \mathcal{L}_A(\hat{\mu}_2^A, \sigma_2^A, 1)$ and not labeling $c_2 := \mathcal{L}_A(\hat{\mu}_2^A, \sigma_2^A, 0)$ in the first period are the same for both, the single-ADR and the cross-ADR debiasing approach. The optimal expected signaling costs for the single-ADR approach is $c_3 := \tilde{V}_S^*$, which is independent of σ_1^{bO} . The optimal signaling cost for our approach is $c(\sigma_1^{bO}) := V_S^*(\rho^{bO}) < c_3$.

The total cost of single-ADR is $\min\{c_1, c_2 + c_3\}$ and for our approach is $\min\{c_1, c_2 + c(\sigma_1^{bO})\}$. The benefit is then defined as $\min\{c_1, c_2 + c_3\} - \min\{c_1, c_2 + c(\sigma_1^{bO})\}$.

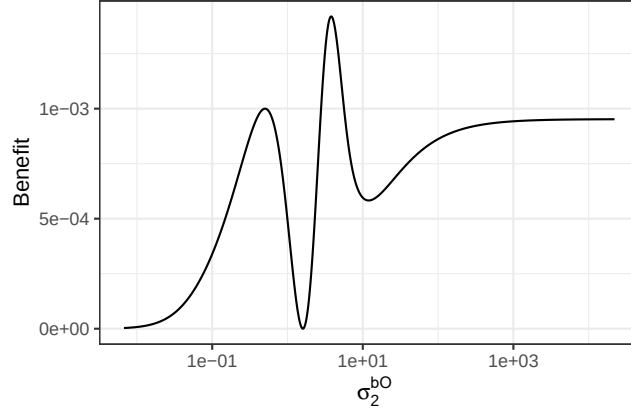
Case: σ_2^S up-down in σ_1^{bO} . Since the signaling costs $c(\sigma_1^{bO})$ is increasing in the posterior variance, it is also up-down in σ_1^{bO} . From this, we note that our approach can result in optimal decisions: wait-and-see | label-immediately | wait-and-see, as σ_1^{bO} increases from 0 to ∞ . We further notice that the label-immediately mid region may or may not exist (*e.g.*, it does not exist whenever $c_1 \geq c_2 + c_3$). Therefore, the benefit is either down-zero-up (if the label-immediately region exists) or down-up in σ_1^{bO} .

Case: σ_2^S down-up in σ_1^{bO} . Since the optimal signaling cost increases in the posterior variance, it is also down-up in σ_1^{bO} . From this, we note that our approach can result in optimal decisions: label-immediately | wait-and-see | label-immediately, as σ_1^{bO} increases from 0 to ∞ . We further notice that the wait-and-see mid region may or may not exist (*e.g.*, it does not exist whenever $c_1 \leq c_2 + c(y^*)$, where y^* the minimizer of $c(\cdot)$). Therefore, the benefit is up-down and potentially zero for extreme values (either small or large) of σ_1^{bO} (this is the case whenever the middle region of wait-and-see exists).

We conclude the benefit is either down-up or up-down in σ_1^{bO} . By the symmetry of the covariance matrices, the comparative static for σ_1^{bC} follows the same pattern as that of σ_1^{bO} .

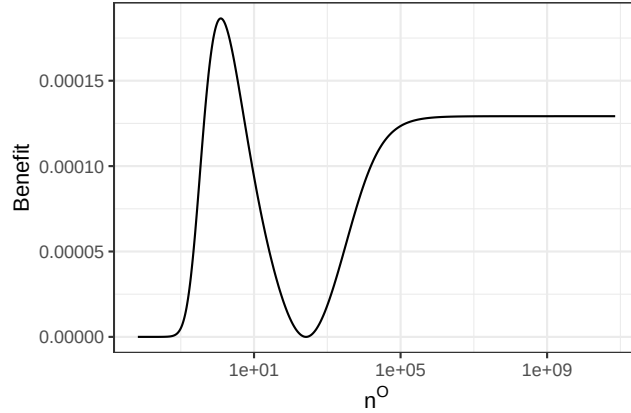
- The benefit has an indeterminate pattern in σ_2^{bO} when we relax [Assumption 2](#). We present a numerical example in [Figure EC.4](#).
- The benefit has an indeterminate pattern in n^O . The increasing behavior characterized in [Theorem 1](#) is no longer true when [Assumption 2](#) is relaxed. We present a numerical example for the lump-sum loss (μ^k -loss function with $k=0$) with $k\hat{\mu}_2^C = 0$ in [Figure EC.5](#). $\square$

Figure EC.4 Example of indeterminate benefit pattern in σ_2^{bO} for a μ^K -loss function with $k = 0$ and $\sigma_1^{bC} \geq 0$ (relaxed **Assumption 2**)



Note. The plot shows the second-period cost difference $\tilde{V}_S^* - V_S^*$ (this is the relevant portion of the benefit whenever $M_{II}^A \approx 0$). The parameters $\hat{\mu}_1^A$ and μ_1^{bO} do not affect the benefit (see **Theorem 1**) and under **Assumption 1**, the parameters ρ^{sC} and ρ^{sO} also do not affect the benefit. We only specify the remaining parameters, namely, $M_1^S = 1$, $M_{II}^S = 0.7$, $k = 0$, $\sigma_2 = 1$, $n^C = 300$, $n^O = 600$, $\mu_2^{bO} = 0$, $\sigma_1^{bO} = 2$, $\rho^{bO} = 0.6$, $\sigma_1^{bC} = 3$, $\sigma_2^{bC} = 2$, $\rho^{bC} = 0.5$, and $\hat{\mu}_2^A = 2.5$.

Figure EC.5 Example of indeterminate benefit pattern in n^O for a μ^k -loss function with $k = 0$ and $\sigma_1^{bC} \geq 0$ (relaxed **Assumption 2**)



Note. The plot shows the second-period cost difference $\tilde{V}_S^* - V_S^*$ (this is the relevant portion of the benefit whenever $M_{II}^A \approx 0$). The parameters $\hat{\mu}_1^C$ and μ_1^{bO} do not affect the benefit, and under **Assumption 1**, the parameters ρ^{sC} and ρ^{sO} also do not affect the benefit. We only specify the remaining parameters, namely, $M_1^S = 1$, $M_{II}^S = 0.9$, $k = 0$, $\sigma_2 = 44$, $n^C = 350$, $\mu_2^{bO} = 0$, $\sigma_1^{bO} = 0.3$, $\sigma_2^{bO} = 0.5$, $\rho^{bO} = 0.7$, $\sigma_1^{bC} = 3$, $\sigma_2^{bC} = 1$, $\rho^{bC} = 0.03$, and $\hat{\mu}_2^A = 0$.

Section E: Misspecified data model

We consider the case in which the key bias-correlation parameter ρ^{bO} is misspecified. Suppose that we incorrectly believe that the observational bias correlation is some $\underline{\rho}^{bO}$ instead of the true ρ^{bO} . This affects the diagonal term in the correlation matrix Σ^{bO} , so $\hat{\mu}^O \sim \mathcal{N}(\mu + \mu^{bO}, \Sigma^{bO} + \Sigma^{sO})$, yet we incorrectly believe the update is instead made with some $\underline{\Sigma}^{bO} \neq \Sigma^{bO}$. We proceed with a Bayesian update under our incorrect assumptions.

$$\begin{aligned} \mu | \hat{\mu}^O, \hat{\mu}^C &\sim \mathcal{N}(\hat{\mu}^S, \Sigma^S), \text{ for} \\ \hat{\mu}^S &:= \Sigma^S ((\Sigma^{sC} + \Sigma^{bC})^{-1} \hat{\mu}^A + (\Sigma^{sO} + \underline{\Sigma}^{bO})^{-1} (\hat{\mu}^O - \mu^{bO})) \text{ and} \\ \Sigma^S &:= ((\Sigma^{sC} + \Sigma^{bC})^{-1} + (\Sigma^{sO} + \underline{\Sigma}^{bO})^{-1})^{-1}. \end{aligned}$$

Similar to Lemma EC.2, we establish the following intermediate result:

LEMMA EC.5. *In the misspecified setting, the true effect size μ and the posterior mean effect size $\hat{\mu}^S$ follow the joint multivariate normal distribution*

$$\begin{bmatrix} \mu \\ \hat{\mu}^S \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} \hat{\mu}^A \\ \hat{\mu}^A \end{bmatrix}, \begin{bmatrix} \Sigma^{sC} + \Sigma^{bC} & \Sigma^S (\Sigma^{sO} + \underline{\Sigma}^{bO})^{-1} (\Sigma^{sC} + \Sigma^{bC}) \\ \Sigma^S (\Sigma^{sO} + \underline{\Sigma}^{bO})^{-1} (\Sigma^{sC} + \Sigma^{bC}) & \Sigma^S (\Sigma^{sO} + \underline{\Sigma}^{bO})^{-1} (\Sigma^{all}) (\Sigma^{sO} + \underline{\Sigma}^{bO})^{-1} \Sigma^S \end{bmatrix} \right),$$

where $\Sigma^{all} := \Sigma^{sC} + \Sigma^{bC} + \Sigma^{sO} + \Sigma^{bO}$, and Σ^S is the misspecified posterior covariance matrix with $\underline{\rho}^{bO}$.

Proof: The proof follows by construction. Recall that data $\hat{\mu}^O = \mu + \mathbf{b}^O + \epsilon^O$, and note that μ and $\mathbf{b}^O + \epsilon^O$ are independent multivariate normal distributions, so we can write their joint distribution as

$$\begin{bmatrix} \mu \\ \mathbf{b}^O + \epsilon^O \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} \hat{\mu}^A \\ \hat{\mu}^{bO} \end{bmatrix}, \begin{bmatrix} \Sigma^{sC} + \Sigma^{bC} & \mathbf{0} \\ \mathbf{0} & \Sigma^{sO} + \Sigma^{bO} \end{bmatrix} \right).$$

It follows that we can write the vector of μ and $\hat{\mu}^S$ as an affine transformation of this multivariate normal distribution:

$$\begin{bmatrix} \mu \\ \hat{\mu}^S \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ \Sigma^S (\Sigma^{sC} + \Sigma^{bC})^{-1} \hat{\mu}^A - \Sigma^S (\Sigma^{sO} + \underline{\Sigma}^{bO})^{-1} \mu^{bO} \end{bmatrix} + \begin{bmatrix} I & \mathbf{0} \\ \Sigma^S (\Sigma^{sO} + \underline{\Sigma}^{bO})^{-1} & \Sigma^S (\Sigma^{sO} + \underline{\Sigma}^{bO})^{-1} \end{bmatrix} \begin{bmatrix} \mu \\ \mathbf{b}^O + \epsilon^O \end{bmatrix}.$$

As a result, it is immediate that μ and $\hat{\mu}^S$ jointly follow a multivariate normal distribution with mean and variance indicated in the statement of the lemma. $\square$

An interesting case of misspecification related to the use of regularization in estimation is the underestimation of the true ρ^{bO} . We can show the following result:

PROPOSITION EC.1. *Under Assumptions 1 and 2 when the true ρ^{bO} is misspecified to instead be some $\underline{\rho}^{bO}$, then $\frac{dV_S^*}{d\rho^{bO}}$ has the opposite sign of ρ^{bO} when evaluated at $\underline{\rho}^{bO} = 0$.*

Proof: For ease of exposition, we consider $\rho^{bO} > 0$, but note that a similar argument can be made for a negative correlation. Using the same notation as in Appendix B.2, the misspecification of the ρ^{bO} parameter does not impact any element of $\Sigma^{bC} + \Sigma^{sC} = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$; note that $a = b = 0$ under Assumptions 1 and 2. However, it does impact the value of the off-diagonal terms of $\Sigma^{bO} + \Sigma^{sO}$, which takes actual value $\Sigma := \begin{bmatrix} d & e \\ e & f \end{bmatrix}$ under the correctly specified ρ^{bO} but is assumed to take value $\underline{\Sigma} := \begin{bmatrix} d & \underline{e} \\ \underline{e} & f \end{bmatrix}$ under the misspecified $\underline{\rho}^{bO}$ value. Further, under Assumptions 1 and 2, $e = \rho^{bO} \sigma_1^{bO} \sigma_2^{bO}$ and $\underline{e} = \underline{\rho}^{bO} \sigma_1^{bO} \sigma_2^{bO}$.

Applying Lemma EC.5, and some careful matrix algebra, yields the joint distribution of the true outcome and the misspecified estimator of the outcome:

$$\begin{aligned} \begin{bmatrix} \mu_2 \\ \hat{\mu}_2^S \end{bmatrix} &\sim \mathcal{N}\left(\begin{bmatrix} \hat{\mu}_2^A \\ \hat{\mu}_2^A \end{bmatrix}, \begin{bmatrix} c & g \\ g & h \end{bmatrix}\right), \text{ for} \\ g &:= \frac{c^2 d}{d(c+f) - \underline{e}^2} \text{ and} \\ h &:= \frac{c^2 d(d(c+f) - \underline{e}(2e - \underline{e}))}{(d(c+f) - \underline{e}^2)^2}. \end{aligned} \quad (\text{EC.18})$$

Note that the value of h depends on the magnitude of the misspecification, *i.e.*, the true e vs. $\underline{e}$.

We next characterize the expected signaling-time loss and its derivative under the misspecified data model:

$$\begin{aligned} V_S^* &= \int \Phi\left(\frac{t_S^* - \hat{\mu}_2^A}{\sqrt{h}}\right) dT_{\text{II}}^S(x) + \int \Phi\left(\frac{x - \hat{\mu}_2^A}{\sqrt{c}}\right) (-dT_{\text{I}}^S(x)) - \int \tilde{\Phi}\left(\frac{x - \hat{\mu}_2^A}{\sqrt{c}}, \frac{t_S^* - \hat{\mu}_2^A}{\sqrt{h}}, \frac{g}{\sqrt{ch}}\right) (dT_{\text{II}}^S(x) - dT_{\text{I}}^S(x)), \text{ so} \\ \left. \frac{dV_S^*}{d\underline{e}} \right|_{\underline{e}=0} &= \phi\left(\frac{t_S^* - \hat{\mu}_2^A}{\sqrt{h}}\right) \cdot \left[\frac{d}{d\underline{e}} \left\{ \frac{t_S^* - \hat{\mu}_2^A}{\sqrt{h}} \right\} \cdot \left(\int_{-\infty}^{\infty} dT_{\text{II}}^S(x) - \int_{-\infty}^{\infty} \Phi\left(\frac{\frac{x - \hat{\mu}_2^A}{\sqrt{c}} - \frac{g}{\sqrt{ch}} \cdot \frac{z\sigma_2^S - \hat{\mu}_2^A}{\sqrt{h}}}{\sqrt{1 - (\frac{g}{\sqrt{ch}})^2}}\right) (dT_{\text{II}}^S(x) - dT_{\text{I}}^S(x)) \right) - \right. \\ &\quad \left. \int_{-\infty}^{\infty} \frac{1}{\sqrt{1 - (\frac{g}{\sqrt{ch}})^2}} \cdot \phi\left(\frac{\frac{x - \hat{\mu}_2^A}{\sqrt{c}} - \frac{g}{\sqrt{ch}} \cdot \frac{z\sigma_2^S - \hat{\mu}_2^A}{\sqrt{h}}}{\sqrt{1 - (\frac{g}{\sqrt{ch}})^2}}\right) \cdot \frac{d}{d\underline{e}} \left\{ \frac{g}{\sqrt{ch}} \right\} (dT_{\text{II}}^S(x) - dT_{\text{I}}^S(x)) \right] \Big|_{\underline{e}=0} \\ &= \phi\left(\frac{t_S^* - \hat{\mu}_2^A}{\sqrt{h}}\right) \cdot \left[\frac{d}{d\underline{e}} \left\{ \frac{t_S^* - \hat{\mu}_2^A}{\sqrt{h}} \right\} \cdot \underbrace{\left(\int_{-\infty}^{\infty} dT_{\text{II}}^S(x) - \int_{-\infty}^{\infty} \Phi\left(\frac{x - t_S^*}{\sigma_2^S}\right) (dT_{\text{II}}^S(x) - dT_{\text{I}}^S(x)) \right)}_{=0} - \right. \\ &\quad \left. \int_{-\infty}^{\infty} \frac{e\sqrt{c}}{d(c+f)\sqrt{f}} \cdot \phi\left(\frac{x - t_S^*}{\sigma_2^S}\right) (dT_{\text{II}}^S(x) - dT_{\text{I}}^S(x)) \right] \Big|_{\underline{e}=0}. \end{aligned}$$

In the derivative, we consider μ^k -loss functions for ease of exposition (similar logic can be applied for general monotone loss functions). Note that the first two terms in the derivative approach zero, as defined by t_S^* (see eq. (EC.7) within Observation 1). The last term will take the sign of $-e$, and since e shares the same sign as ρ^{bO} , this completes the proof. $\square$

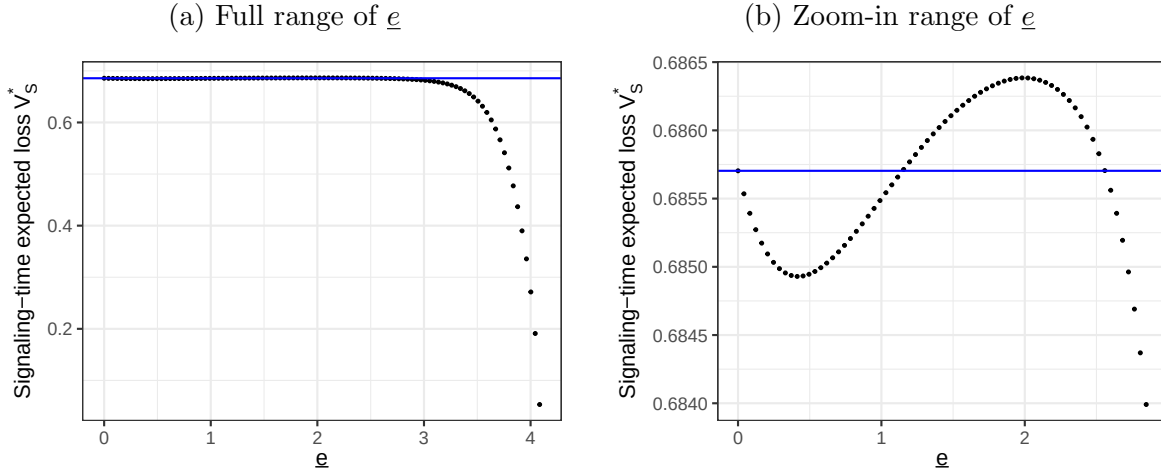
This result shows that cross-ADR yields lower expected signaling costs under a misspecified data model that assumes small bias correlation (*i.e.*, $\rho^{bO} \rightarrow 0$, which is the case when ρ^{bO} is estimated under strong regularization). In the main paper, we have also shown that the cross-ADR approach is better when correctly specified (*i.e.*, $\rho^{bO} \rightarrow \rho^{bO}$). However, we cannot guarantee its superiority under an intermediate level of misspecification. We illustrate this rather uncommon, yet plausible, case with a numerical example.

We generated a random set of 400,000 problem instances satisfying Assumptions 1 and 2. We consider a lump-sum loss function, where the costs M_{I} and M_{II} are incurred for any Type I and II errors (regardless of magnitude). Here, we know that $t_S^* = z\sigma_2^S$ for $z = \Phi^{-1}\left(\frac{M_{\text{I}}}{M_{\text{I}} + M_{\text{II}}}\right)$. This yields the following expected loss:

$$V_S^* = M_{\text{II}} \cdot \Phi\left(\frac{z\sigma_2^S - \hat{\mu}_2^A}{\sqrt{h}}\right) + M_{\text{I}} \cdot \Phi\left(\frac{-\hat{\mu}_2^A}{\sqrt{c}}\right) - (M_{\text{I}} + M_{\text{II}}) \cdot \tilde{\Phi}\left(\frac{-\hat{\mu}_2^A}{\sqrt{c}}, \frac{z\sigma_2^S - \hat{\mu}_2^A}{\sqrt{h}}, \frac{g}{\sqrt{ch}}\right). \quad (\text{EC.19})$$

Figure EC.6 displays V_S^* for a particular problem instance. From Figure EC.6 (a), it is apparent that the expected loss is approximately constant for $\underline{e} \in [0, 3]$ before dropping sharply for $\underline{e} \in [3, e]$. However, zooming in on the left side of the plot (Figure EC.6 (b)), we see that, in fact, the expected loss for $\underline{e} \in [1.16, 2.55]$ is larger than the expected loss of the single-ADR model with $\underline{e} = 0$ (blue horizontal line). The difference is quite small (the increased cost takes a maximum value of 0.00068, or approximately 0.1% of the expected loss under the single-ADR model). None of the 400,000 instances featured a *managerially significant* benefit for the single-ADR approach for any $0 < \rho^{bO} < \rho^{bO}$.

Figure EC.6 Example of expected signaling-time loss for the cross-ADR approach being higher than expected signaling-time loss for the single-ADR approach under misspecification in ρ^{bO} .



Note: Blue horizontal line corresponds to the single-ADR expected loss. Parameter values: $c = 1.34877$, $d = 1.680853$, $f = 9.936704$, $e = 0.9995541\sqrt{df} = 4.085$, $\hat{\mu}_2^A = 0.5693184$, $M_I = 10.45653$, and $M_{II} = 1$.

Section F: Heterogeneous treatment effects

We present a stylized generative model of our data inputs and use it to examine the impact of heterogeneous treatment effects and the link to our baseline model. We consider a single drug and ADR of interest, and for ease of exposition, we measure treatment effect sizes as log risk ratios ($\log(\text{RR})$), but note that a similar analysis can be done using log odds ratio ($\log(\text{OR})$).

Our generative model splits up patients into groups based on two dimensions:

1. *ADR frequency*: We define a high-type (H) and a low-type (L) that differ in two dimensions. First, the groups differ in the baseline rate of the ADR (here, the “baseline rate” of the ADR is the ADR rate without taking the drug of interest; assume $p_H \geq p_L$).

One might think of H as a subpopulation with some comorbid condition (*e.g.*, diabetes) that boosts their risk for the ADR of interest. We assume individuals with the ADR of interest always report it in observational data.⁴ Second, we allow our groups to differ in the reporting rate of *any other* side effects. Formally, α_L (α_H) is defined as the proportion of the population that does not have the ADR of interest, but that will report *some other* side effect in the L (H) group. For simplicity, we assume the same reporting proportion regardless of whether individuals take the drug. The H vs. L differentiation does not capture heterogeneous effects (which would imply that the drug impacts different patients differently)—the drug should have the same effect size for the ADR for types H and L.

2. *Heterogeneous effects*: We define groups A and B, which experience *heterogeneous effects* of the drug on the ADR. That is, the effect size for the ADR differs under group A vs. B (without loss of generality, we assume effect sizes satisfying $\mu_A < \mu_B$, meaning B experiences a more harmful effect than A).

⁴ This assumption is made to simplify the exposition and can be easily relaxed to model under-reporting of the focal ADR.

With this setup in place, we have four subpopulation types:

We have two settings s , either clinical trial ($s = C$) or observational data ($s = O$). In both of these settings, some individuals will take the drug of interest ($t = 1$) or a standard therapy ($t = 0$). Let n_{st}^i indicate the observed number of patients of type $i \in \{AH, AL, BH, BL\}$ taking therapy $t \in \{0, 1\}$ in setting $s \in \{C, O\}$, let N_t^i refer to the (unobserved) overall patient population of type i who choose treatment t , and let $N^i := N_0^i + N_1^i$ refer to the (unobserved) overall patient population of type i . We analyze the case without heterogeneous treatment effects in subsection F.1 and show that imbalances in treatment uptake rates between patients with different baseline ADR rates and/or reporting rates (the L and H groups) can produce observational bias. In subsection F.2, we examine the case with heterogeneous treatment effects and show that clinical trial generalization bias, which is part of our baseline model, can be thought of as a direct result of heterogeneous treatment effects.

F.1. No heterogeneous effect – only group A

Here we model that group B does not exist and that we use the $\log(\text{RR})$ as our outcome measure.⁵ We examine the estimated effect sizes under our model, given the underlying data-generating process.

F.1.1. Clinical trial data. Consider first a clinical trial that draws $n_{C0}^{AL} = n_{C1}^{AL} := n^{AL}$ participants of type AL into each of the control and treatment arm (via 50/50 randomization) along with $n_{C0}^{AH} = n_{C1}^{AH} := n^{AH}$ participants of type AH into each arm.⁶ Notice that n^{AL} and n^{AH} have been defined arbitrarily and do not need to match the population distribution of L-type and H-type patients (the N^{AL} vs. N^{AH} breakdown). Since clinical trials ascertain *all side effects* that occur, we have:

$$\begin{aligned} \text{Focal ADR event rate in control arm} &= \frac{n^{AL}p_L + n^{AH}p_H}{n^{AL} + n^{AH}}, \\ \text{Focal ADR event rate in treatment arm} &= \frac{n^{AL}p_L e^{\mu_A} + n^{AH}p_H e^{\mu_A}}{n^{AL} + n^{AH}}, \text{ and} \\ \text{Clinical trial } \log(\text{RR}) &:= \log \left(\frac{\text{Focal ADR event rate in treatment arm}}{\text{Focal ADR event rate in control arm}} \right) = \mu_A. \end{aligned}$$

In this setting, the clinical trial’s randomization ensures that the estimate is not affected by generalization bias even if the trial does not sample types L and H at the rate at which they appear in the population (notice that the result would continue to hold even if we randomized with some rate other than 50/50). The lack of generalization bias is a direct consequence of not having effect size heterogeneity (all patients are in group A with effect size μ_A).

⁵ We use log risk ratio to simplify the exposition. Note that the log risk ratio and log odds ratio are approximately equal for rare ADRs.

⁶ Often, clinical trials stratify on health status indicators, so the assumption here of near-perfect balance between the control and treatment arm in both the L and H types is not unreasonable.

Table EC.1 Population groups

Population type	ADR reporting rate	
	Focal ADR	Other ADRs
AH	$p_H e^{\mu_A}$	$\alpha_H(1 - p_H e^{\mu_A})$
AL	$p_L e^{\mu_A}$	$\alpha_L(1 - p_L e^{\mu_A})$
BH	$p_H e^{\mu_B}$	$\alpha_H(1 - p_H e^{\mu_B})$
BL	$p_L e^{\mu_B}$	$\alpha_L(1 - p_L e^{\mu_B})$

F.1.2. Observational data. We have four patient counts: n_{O0}^{AL} , n_{O1}^{AL} , n_{O0}^{AH} , and n_{O1}^{AH} , which we should interpret as the counts of patients of each type choosing to take the new treatment (or not) and reporting some side effect (either the focal ADR or some other ADR). We have the following expected counts:

$$\begin{aligned} n_{O0}^{AL} &= N_0^{AL}(p_L + (1 - p_L)\alpha_L), & n_{O1}^{AL} &= N_1^{AL}(p_L e^{\mu_A} + (1 - p_L e^{\mu_A})\alpha_L), \\ n_{O0}^{AH} &= N_0^{AH}(p_H + (1 - p_H)\alpha_H), & n_{O1}^{AH} &= N_1^{AH}(p_H e^{\mu_A} + (1 - p_H e^{\mu_A})\alpha_H). \end{aligned}$$

Note that the counts include all patient reports; that is, they include those who experience the side effect of interest and those who did not but still report some other side effect; FAERS does not contain information about individuals who did not report any side effects. We define the proportion of reports corresponding to our ADR of interest in each subgroup as follows:

$$\begin{aligned} p_0^{AL} &= \frac{p_L}{p_L + (1 - p_L)\alpha_L}, & p_1^{AL} &= \frac{p_L e^{\mu_A}}{p_L e^{\mu_A} + (1 - p_L e^{\mu_A})\alpha_L}, \\ p_0^{AH} &= \frac{p_H}{p_H + (1 - p_H)\alpha_H}, & p_1^{AH} &= \frac{p_H e^{\mu_A}}{p_H e^{\mu_A} + (1 - p_H e^{\mu_A})\alpha_H}. \end{aligned}$$

Then we have:

$$\begin{aligned} \text{Focal ADR event rate for patients receiving } t=0 &= \frac{n_{O0}^{AL} p_0^{AL} + n_{O0}^{AH} p_0^{AH}}{n_{O0}^{AL} + n_{O0}^{AH}}, \text{ and} \\ \text{Focal ADR event rate for patients receiving } t=1 &= \frac{n_{O1}^{AL} p_1^{AL} + n_{O1}^{AH} p_1^{AH}}{n_{O1}^{AL} + n_{O1}^{AH}} \end{aligned}$$

$$\begin{aligned} \text{Obs. log(RR)} &:= \log \left(\frac{\text{Focal ADR event rate for patients receiving } t=1}{\text{Focal ADR event rate for patients receiving } t=0} \right) \\ &= \mu_A + \underbrace{\log \left(\frac{N_1^{AL} p_L + N_1^{AH} p_H}{N_0^{AL} p_L + N_0^{AH} p_H} \frac{N_0^{AL}(p_L + (1 - p_L)\alpha_L) + N_0^{AH}(p_H + (1 - p_H)\alpha_H)}{N_1^{AL}(p_L e^{\mu_A} + (1 - p_L e^{\mu_A})\alpha_L) + N_1^{AH}(p_H e^{\mu_A} + (1 - p_H e^{\mu_A})\alpha_H)} \right)}_{\text{Observational data bias}}. \end{aligned} \tag{EC.20}$$

In a perfectly homogeneous population, with equal ADR frequency rate ($p_L = p_H \equiv p$), reporting patterns ($\alpha_L = \alpha_H \equiv \alpha$), and uptake in each group ($\frac{N_1^{AL}}{N_0^{AL}} = \frac{N_1^{AH}}{N_0^{AH}}$), the observational bias is expected to be small for rare enough ADRs (see [Figure EC.7](#)):

$$\text{Observational data bias} = \log \left(\frac{N_0^{AL} p + N_0^{AH} (1 - p) \alpha}{N_0^{AL} p e^{\mu_A} + N_0^{AH} (1 - p e^{\mu_A}) \alpha} \right) \rightarrow 0 \text{ as } p \rightarrow 0.$$

More generally, we expect to see observational bias be a composite of having different ADR rates and reporting patterns in the population. We isolate each effect as follows:

Heterogeneity in baseline rates only: We consider the special case where no reporting differences exist between the L and H-type ($\alpha_L = \alpha_H \equiv \alpha$), but the baseline rate is different with $p_H > p_L$. Of course, the core premise of our work is that (unlike in an RCT) the ratios N_1^{AL}/N_0^{AL} and N_1^{AH}/N_0^{AH} can differ significantly from one another, capturing that the L and H types may have different uptake of the new treatment. We can formalize this by defining:

$$\begin{aligned} \text{log(OR) of AH vs. AL uptake} &:= \log \left(\frac{\text{odds of AH choosing } t=1}{\text{odds of AL choosing } t=1} \right) \\ &= \log \left(\frac{N_1^{AH}/N_0^{AH}}{N_1^{AL}/N_0^{AL}} \right) \equiv \kappa. \end{aligned}$$

We illustrate the effect of $\kappa \neq 0$ with a simple numerical example. Consider $\mu_A = 0.34$ and $p_H = 0.4$, population size in the AH group that is some γ times the population size in the AL group (*i.e.*, $N_0^{AH} + N_1^{AH} = \gamma(N_0^{AL} + N_1^{AL})$), and equal distribution of the AL group between treatment options ($N_0^{AL} = N_1^{AL}$). We vary the degree of heterogeneity in terms of the ADR baseline rate $p_L/p_H \in (0, 1]$. Figure EC.8(a) shows the observational log risk ratio calculated with eq. (EC.20). We note that when the H-type is more likely to receive treatment ($\kappa > 0$), we tend to observe a positive bias, and the effect is more pronounced when the H-type has a much higher baseline rate relative to the L-type ($p_L/p_H \ll 1$). We also see that even in a perfectly homogeneous population ($p_L/p_H = 1$), bias will persist due to misreporting ($\alpha < 1$) and the non-negligible baseline rate ($p_H \gg 0$), resulting in an underestimation of the true effect size.

Heterogeneity in reporting patterns only: Similar to above, we examine the impact of differential reporting patterns on the observational log risk ratio, assuming that the H and L-type groups are similar in terms of their baseline risk (*i.e.*, $p_H = p_L = p$). Figure EC.8(b) shows the observational effect size for varying reporting rates ratio α_H/α_L and for different values of κ , which implies different AH vs. AL uptake. We note that even when uptake is the same in the H and L groups ($\kappa = 0$), we will observe some bias due to underreporting $\alpha_H < \alpha_L < 1$. Underreporting from individuals not experiencing the ADR of interest results in an inflated focal ADR event rate for both the treatment and no-treatment groups, but the latter effect is larger since the portion of patients without the ADR is larger in the non-treatment group (*i.e.*, $(1 - p) > (1 - pe^{\mu_A})$).

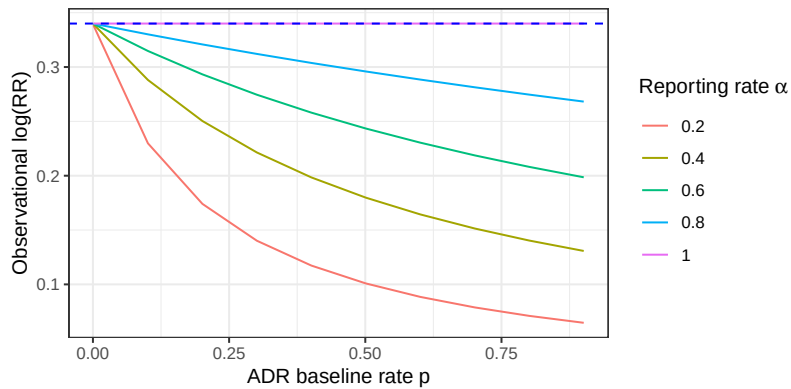
F.2. Heterogeneous effects – groups A and B

Now we move to the setting with groups A and B with heterogeneous treatment effects $\mu_B > \mu_A$.

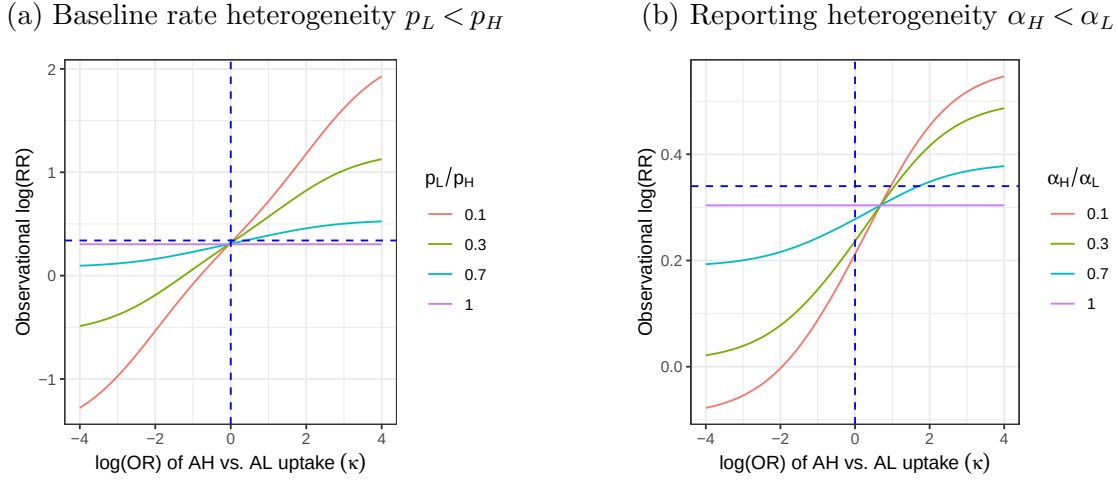
F.2.1. Clinical trial data. Since we have two groups with different effect sizes, we expect the proportion from A vs. B to impact the average treatment effect in the trial, and we further expect that the trial oversampling one of the two groups will result in *generalization bias*. Assume as before that our trial distributes n^{AL} , n^{AH} , n^{BL} , and n^{BH} patients to both the control and treatment arms. Then we have:

$$\text{Event rate in control arm} = \frac{n^{AL}p_L + n^{AH}p_H + n^{BL}p_L + n^{BH}p_H}{n^{AL} + n^{AH} + n^{BL} + n^{BH}},$$

Figure EC.7 Observational log(RR) for homogeneous population



Note. $\mu_A = 0.34$.

Figure EC.8 Observational effect size for varying log(OR) of AH vs. AL uptake

Note: (a) $\mu_A = 0.34$, $p_H = 0.4$, $\alpha_L = \alpha_H = 0.8$. (b) $\mu_A = 0.34$, $p_H = p_L = 0.4$, $\alpha_L = 0.8$. We set $\gamma = 1$; the results are robust to this value. Note the different y-axis scales in the two figures.

$$\text{Event rate in treatment arm} = \frac{n^{AL}p_L e^{\mu_A} + n^{AH}p_H e^{\mu_A} + n^{BL}p_L e^{\mu_B} + n^{BH}p_H e^{\mu_B}}{n^{AL} + n^{AH} + n^{BL} + n^{BH}}, \text{ and}$$

$$\text{Clinical trial log(RR)} = \log(\beta e^{\mu_A} + (1 - \beta)e^{\mu_B}), \text{ for}$$

$$\beta := \frac{n^{AL}p_L + n^{AH}p_H}{n^{AL}p_L + n^{AH}p_H + n^{BL}p_L + n^{BH}p_H}.$$

Here, we can interpret β to indicate the expected proportion of all events in the control arm that come from group A. If the population rates of groups AL, AH, BL, and BH induced some β^* , then we have positive generalization bias for clinical trials with $\beta < \beta^*$ and negative generalization bias for clinical trials with $\beta > \beta^*$. Thus, with heterogeneous treatment effects, the clinical trial estimates can be biased, which in our model is captured through the generalization bias parameters $\mathbf{b}^C \sim \mathcal{N}(\boldsymbol{\mu}^{b^C}, \boldsymbol{\Sigma}^{b^C})$.

F.2.2. Observational data. Using the notation introduced in subsection F.1.2 for the observational event counts and rates ($n_{O_i}^i$ and p_i^i for $i \in \{AL, AH, BL, BH\}$), we define

$$\text{Event rate for patients receiving } t = 0 = \frac{n_{O_0}^{AL}p_0^{AL} + n_{O_0}^{AH}p_0^{AH} + n_{O_0}^{BL}p_0^{BL} + n_{O_0}^{BH}p_0^{BH}}{n_{O_0}^{AL} + n_{O_0}^{AH} + n_{O_0}^{BL} + n_{O_0}^{BH}},$$

$$\text{Event rate for patients receiving } t = 1 = \frac{n_{O_1}^{AL}p_1^{AL} + n_{O_1}^{AH}p_1^{AH} + n_{O_1}^{BL}p_1^{BL} + n_{O_1}^{BH}p_1^{BH}}{n_{O_1}^{AL} + n_{O_1}^{AH} + n_{O_1}^{BL} + n_{O_1}^{BH}}, \text{ and}$$

$$\text{Observational log(RR)} = \log(\beta e^{\mu_A} + (1 - \beta)e^{\mu_B} + C), \text{ for}$$

$$\beta := \frac{N_0^{AL}p_L + N_0^{AH}p_H}{N_0^{AL}p_L + N_0^{AH}p_H + N_0^{BL}p_L + N_0^{BH}p_H},$$

$$C := \frac{e^{\mu_A}((N_1^{AL}m - N_0^{AL})p_L + (N_1^{AH}m - N_0^{AH})p_H) + e^{\mu_B}((N_1^{BL}m - N_0^{BL})p_L + (N_1^{BH}m - N_0^{BH})p_H)}{N_0^{AL}p_L + N_0^{AH}p_H + N_0^{BL}p_L + N_0^{BH}p_H}, \text{ and}$$

$$m := \frac{n_{O_0}^{AL} + n_{O_0}^{AH} + n_{O_0}^{BL} + n_{O_0}^{BH}}{n_{O_1}^{AL} + n_{O_1}^{AH} + n_{O_1}^{BL} + n_{O_1}^{BH}}.$$

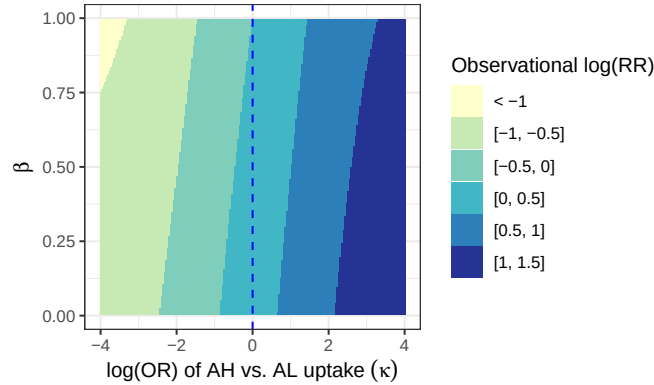
We see that the key difference from the clinical trial setting is that there is a C term added to the interior of the log(RR). This term takes value 0 if the proportion of each of the four groups choosing treatment is the same (as would be the case in a clinical trial). It otherwise tends to take a positive value when groups B and H disproportionately select treatment, while it tends to take a negative value when groups A and L disproportionately select treatment.

Similar to the homogeneous effect results in subsection F.1, we can get significant observational bias by varying the treatment rates between our treatment groups. To illustrate this, we compute the $\log(\text{RR})$ under various allocations. Note that there are 8 N_t^i values (the number with $t = 1$ and $t = 0$ for each of our four groups). These are fully specified with the following assumptions, designed to match our Group A-only setup as closely as possible:

- Noting that γ played a second-order role in the setting with Group A-only, we assume for simplicity that $\gamma = 1$ for both groups, meaning $N_0^{AH} + N_1^{AH} = N_0^{AL} + N_1^{AL}$ and $N_0^{BH} + N_1^{BH} = N_0^{BL} + N_1^{BL}$.
- Noting that the $\log(\text{OR})$ of AH vs. AL uptake and/or reporting (κ) played a crucial role in the observational bias for the setting with Group A-only, we vary its value under the assumption that the value is the same for groups A and B: $\log \frac{N_1^{AH}/N_0^{AH}}{N_1^{AL}/N_0^{AL}} = \log \frac{N_1^{BH}/N_0^{BH}}{N_1^{BL}/N_0^{BL}} = \kappa$.
- As in the previous numerical example, we assume $N_0^{AL} = N_1^{AL}$ and $N_0^{BL} = N_1^{BL}$.
- Noting the critical role that the balance between A and B plays in the clinical trial generalization bias, we vary β as defined above.

Figure EC.9 captures how the observational $\log(\text{RR})$ varies as we change our parameters of κ and β .

Figure EC.9 Observational $\log(\text{RR})$ for varying $\log(\text{OR})$ of AH vs. AL uptake and/or reporting (x-axis) and generalization bias (y-axis)



Note. $p_L = 0.1$, $p_H = 0.4$, $\alpha_L = 0.8$, $\alpha_H = 0.5$, $\mu_A = 0$, $\mu_B = 0.34$, and $\gamma = 1$.

As expected, both β (which controls the A vs. B mix) and κ (which controls the L vs. H mix) play a role in the bias we observe. However, at least for our parameterization here, κ seems to play a much larger role, indicating that we likely see much larger bias magnitudes in the observational data compared to the clinical trial data. Indeed, note that the clinical trial $\log(\text{RR})$ is bounded between $\mu_A = 0$ and $\mu_B = 0.34$ (this is $\kappa = 0$ in the heatmap), while the effect sizes from observational data can vary far more widely than that.

Appendix References

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